

Isaac Computer Science GCSE Book

Isaac CS Team

Contents

1	Network fundamentals	1
2	The internet	5
3	Network hardware	9
4	Boolean logic	11
5	Systems architecture	19
6	Memory and storage	27
7	Hardware	37
8	Software	43
9	Operating systems	47
10	High- and low-level languages	53
11	Translators	59
12	Social engineering	62
13	Malicious software	67
14	Network security	70
15	Representation of numbers	76
16	Representation of text	89
17	Representation of images	95
18	Representation of sound	100
19	Compression	106
20	Encryption	111
21	Database concepts	114
22	SQL	117
23	Data structures	126
24	Searching algorithms	132
25	Sorting algorithms	145
26	Legislation	151
27	Impacts of technology	156
28	Programming concepts	158

29	String handling	172
30	Subroutines	178
31	File handling	186
32	Integrated development environments (IDEs)	194
33	Procedural programming	198
34	Object-oriented programming	200
35	Design	205
36	Testing	213
37	Computational thinking	216

1 Network fundamentals

A **network** is a group of devices connected together, wirelessly or using cables, so that they can communicate with each other and share resources. Networks can be structured in different ways (called **topologies**), and may also be categorised as **client-server** or **peer-to-peer networks**. There are benefits and drawbacks to each approach.

Network **standards and protocols** allow a wide range of devices from different manufacturers to communicate with each other. The speed at which a network operates is important to most users and you will learn about the factors that impact a network's **performance**.

1.1 Star network topologies

AQA

EDEXCEL

EDUQAS

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WJEC

isaacacs.org/bq/gcse_net_02

► A star network topology is where all the nodes are arranged in a star layout and connected to a device at its centre point. What is the name of the device at the centre of the star which can direct the signals to their intended recipients only?

1.2 Client-server and peer-to-peer networking

OCR

isaacacs.org/bq/gcse_net_11

► Below are statements that highlight some of the differences between client-server and peer-to-peer networking. Pick one statement that is **not** correct.

- ☐ In a peer-to-peer network, any device can share resources with any other device in the network.
- ☐ In a peer-to-peer network, a computer can act as both a client and a server.
- ☐ Management of network security and maintenance is usually easier in a client-server model.
- ☐ In a client-server model, all processing must be done in the server side.

1.3 Network topologies

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isaacss.org/bq/gcse_net_01

A local hospital has a computer network that uses a bus network topology. It is finding that the network is slow and is taking too long to retrieve patient records. They have been advised to create a mesh network.

- For which of the following reasons is a mesh network the correct choice?
- ☐ All workstations are connected to a central point, like a hub or a switch.
 - ☐ Mesh networks are cheap to set up and data is transmitted very quickly.
 - ☐ Maintenance costs are lower on a mesh network, because they don't use terminators.
 - ☐ Mesh networks are the most reliable type of network, because data can take a number of possible routes to get to the intended recipient.

1.4 Peer-to-peer networking

OCR

isaacss.org/bq/gcse_net_12

- Which one of the statements below correctly describes the peer-to-peer networking model?
- ☐ The peer-to-peer networking model is very secure, because every device goes through an authentication process before it joins the network.
 - ☐ In peer-to-peer networking, devices can't share their resources with, and request resources from the other devices in the network.
 - ☐ In peer-to-peer networking, there is no central server.
 - ☐ Peer-to-peer networking involves a central, powerful computer called a server, and a number of client computers that need to connect to the server to carry out specific tasks.

1.5 Ethernet

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isaacscs.org/bq/gcse_net_17

- Which of the following statements about Ethernet is **not** correct?
- ☐ All Ethernet devices can process the data received from the router, because they all use the same protocols.
 - ☐ Ethernet is a method to connect computers to a network using a wireless router.
 - ☐ Ethernet is a group of networking technologies commonly used in local area networks.
 - ☐ Ethernet routers use the same set of protocols.

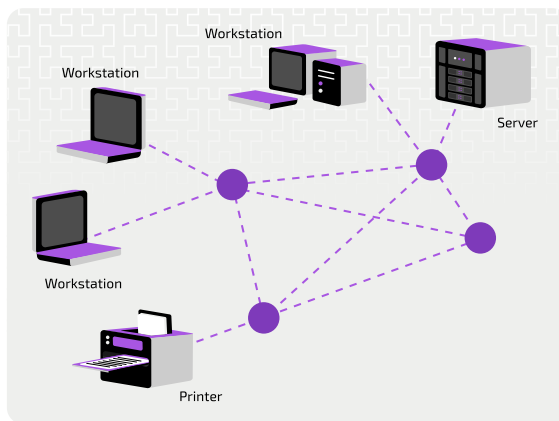
1.6 Shape of the network 1

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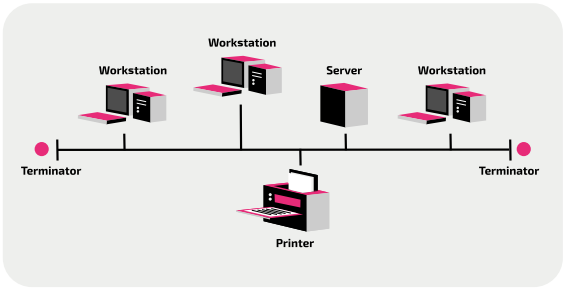
isaacscs.org/bq/net_network_g01

- What topology is illustrated in the diagram?

1.7 Shape of the network 2

AQA EDEXCEL EDUQAS WJEC

isaacacs.org/bq/net_network_g02

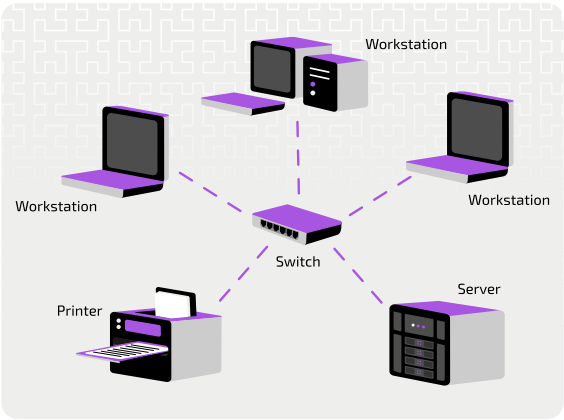


► What topology is illustrated in the diagram?

1.8 Shape of the network 3

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/net_network_g03



► What topology is illustrated in the diagram?

2 The internet

The **internet** is a global system of interconnected computer networks. The internet facilitates a wide range of services, including the World Wide Web, email, internet telephony, and file sharing. In order to communicate to do any of those things, computers must use a common set of protocols referred to as the **TCP/IP stack**.

At a more advanced level, it is important to understand that **standards** (as well as **protocols**) are essential for the internet to function. JSON and XML are both standards for the interchange of data and provide a standard for data exchange via the World Wide Web.

2.1 HTTP protocol

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isaaccs.org/bq/gcse_net_05

► Networks use protocols to establish a common set of rules to communicate. Network communication on the internet is arranged in abstract layers called the TCP/IP model (also referred to as the TCP/IP stack). At which layer in this model would you expect to see the HTTP protocol?

- ☐ Link or data link layer
- ☐ Transport layer
- ☐ Application layer
- ☐ Network or internet layer

2.2 TCP protocol

AQA

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WJEC

isaaccs.org/bq/gcse_net_06

► Networks use protocols to establish a common set of rules to communicate. Network communication on the internet is arranged in abstract layers called the TCP/IP model (also referred to as the TCP/IP stack). At which layer in this model would you expect to see the TCP protocol?

- ☐ Transport layer
- ☐ Application layer
- ☐ Network or internet layer
- ☐ Link or data link layer

2.3 Data packets

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isaacacs.org/bq/gcse_net_10

When data is sent across a network it is broken into packets, each with their own control information.

► Which of the following pieces of data are not part of a data packet?

- ☐ Sender's IP address
- ☐ Receiver's IP address
- ☐ Time the data was sent
- ☐ Checksum

2.4 Dude, where's my page?

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EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_net_13

► Number the following statements in order, to explain what happens when you type the URL of a webpage into your browser. (Assume that none of the information is cached.)

- ☐ The DNS server sends your computer the relevant IP address.
- ☐ The web server sends your computer the webpage.
- ☐ The DNS server looks up the IP address of the server that you want.
- ☐ You type the URL into your browser.
- ☐ Your computer sends a request to the web 'servers IP address.
- ☐ Your computer sends a lookup request to the DNS server.

2.5 Layers of protocols

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isaacscs.org/bq/gcse_net_14

The TCP/IP networking model is split into four layers. Each of these layers has a specific purpose and supports different protocols (rules for communication). The four layers are:

1. Application layer
2. Transport layer
3. Network layer (internet layer)
4. Link layer (data link layer)

Below are four commonly used communication protocols. Each protocol works on only a single layer of the layers listed above.

► Using the order of the layers given above, number the protocols so that they are in the same position in the list as the layer that they work on:

☐

UDP

☐

IP

☐

Wi-Fi

☐

HTTP

2.6 IP addresses

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/net_internet_g01

► Why do all devices that connect to a network or the internet need a unique IP address?

- ☐ To provide the device with a particular identifier
- ☐ It reduces the reliance on the transport layer
- ☐ It increases the speed of transmission

2.7 Protocols

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/net_internet_g02

- What best describes an internet protocol?
- ☐ A set of rules that enables two or more digital devices to communicate
 - ☐ Hardware that is used by DNS servers
 - ☐ Software that enables a digital device to access the internet

2.8 Layers and protocols

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/net_internet_g03

► Complete the table by adding the correct layer or protocol into the spaces so that each protocol matches the layer at which it operates.

Layer	Protocol
Application	_____
_____	TCP
Internet	_____
_____	Ethernet

- ☐ Frame
- ☐ CAT4
- ☐ Link
- ☐ IP
- ☐ Packet
- ☐ Wireless
- ☐ Transport
- ☐ HTML
- ☐ HTTP

3 Network hardware

Network hardware is the term used for all of the hardware components that can be found in computer networks. This includes the **network interface controllers**, that are used to connect individual devices to a network, and components such as **switches** and **routers**, that are used to build the backbone of the network.

Devices can be connected by cable or wirelessly. Many networks have both wired and wireless connections.

3.1 Network problem

EDUQAS OCR

isaacscs.org/bq/net_hard_g01

► An international fashion brand with shops and offices in different cities has many interconnected networks. In one of the offices there is a hardware fault and none of the computers can communicate with each other. Devices in other shops and offices can still communicate with each other. Which option is the most likely hardware fault?

- ☐ A switch connecting computers in the affected office has broken.
- ☐ The router in the affected office has been switched off.
- ☐ The NICs have lost their configuration settings.
- ☐ The modem has broken.

3.2 Network media

AQA EDEXCEL OCR

isaacscs.org/bq/net_hard_g02

► Which of the following network media can provide reliable communication over a long distance?

- ☐ Coaxial cable
- ☐ Wi-Fi
- ☐ UTP (Ethernet) cable
- ☐ Fibre-optic cable

3.3 Your data packet

OCR

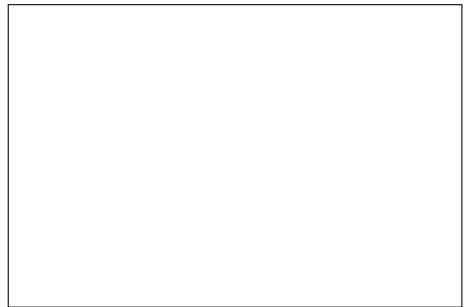
isaaccs.org/bq/net_hard_g03

You are researching universities using a school computer which is connected to the school's LAN. You open a web browser and type the website address of a university into the address bar and press the Enter key. This creates a data packet.

► Put the following network hardware components in the order in which they would receive the data packet containing this request.

(**Note:** This is not an exhaustive list of all hardware required. Some steps in the journey have been left out.)

- ☐ Router
- ☐ Fibre-optic cable
- ☐ Network interface card
- ☐ Switch



4 Boolean logic

Computers are made up of digital circuits. Each component in a circuit has an input voltage that can be evaluated as **True or False**, or **1 or 0**. **Boolean logic** is a way of describing the circuits inside a computer, and the same concept can be represented by a **logic circuit diagram**, a **truth table**, or a **Boolean expression**.

At a more advanced level, you will study more complex logic gates, although all circuits can ultimately be derived from the combination of a small subset of basic gates. Using a formal notation for describing the combination of logical operations will allow you to apply standard rules to simplify complex Boolean expressions. You will explore some standard circuits in detail, such as a **full adder**, which allows arithmetic addition, and a **D-type flip-flop** circuit, which is a very basic unit of memory.

4.1 Find the output

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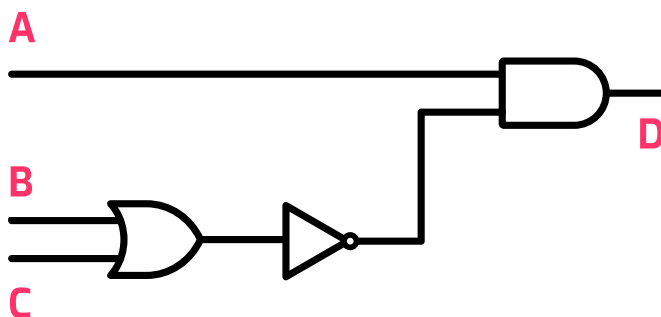
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isaaccs.org/bq/gcse_bool_02



► Is the output (D) of the circuit above T or F when all its inputs (A, B, C) are T?

Answer:

4.2 Which gate is it?

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_bool_08

► Which logic gate does the truth table below represent?

INPUT	INPUT	OUTPUT
0	0	1
0	1	0
1	0	0
1	1	0

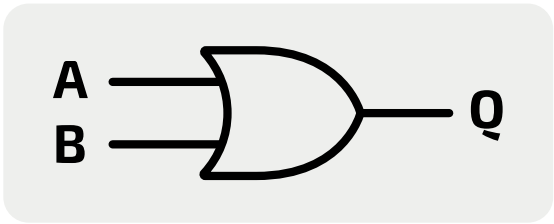
- ☐ NOR
- ☐ NOT
- ☐ NAND
- ☐ XOR

4.3 Choose a truth table

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_bool_01

► Circle the correct truth table for the logic gate represented by the following symbol.



A	B	OUTPUT
0	0	0
0	1	1
1	0	1
1	1	1

A	B	OUTPUT
0	0	0
0	1	1
1	0	1
1	1	0

A	B	OUTPUT
0	0	0
0	1	0
1	0	0
1	1	1

A	OUTPUT
0	1
1	0

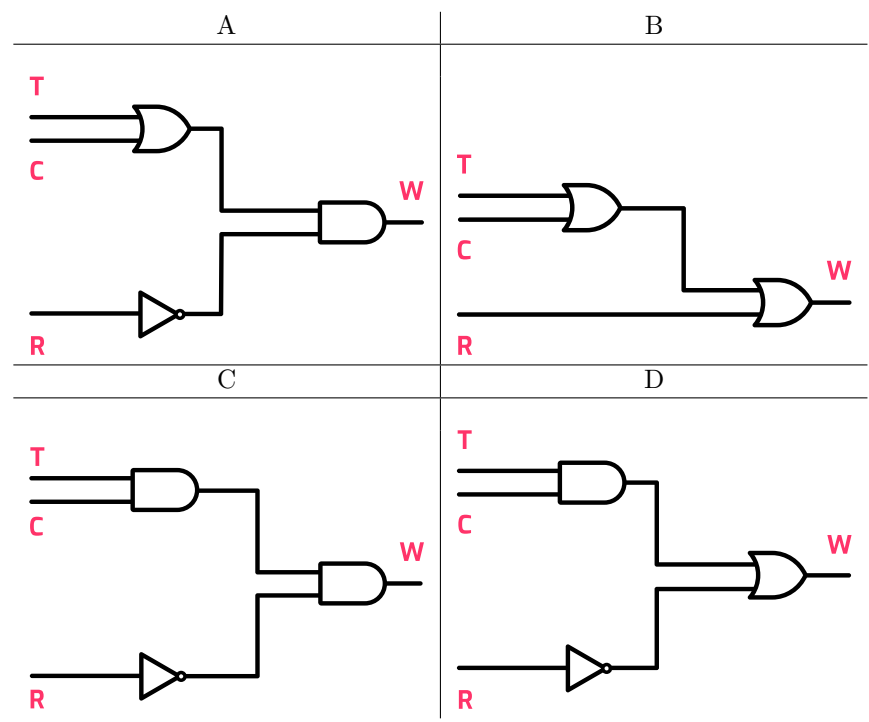
4.4 Choose a logic circuit diagram

AQA OCR

isaaccs.org/bq/gcse_bool_03

A watering sprinkler (W) is on when the timer (T) is set, the temperature (C) exceeds 26° Celsius, and the rain sensor (R) does not detect rainfall.

► Select the correct logic diagram for the sprinkler.



Answer:

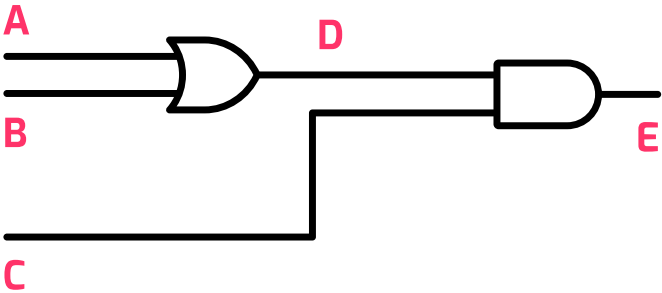
4.5 Find truth table values

AQA OCR

isaacss.org/bq/gcse_bool_04

► Fill in the missing values in the truth table for the logic circuit diagram below, given the following inputs:

A	B	C	D	E
0	0	0	0	0
0	0	1	0	0
0	1	0	1	0
0	1	1	1	1
1	0	0		0
1	0	1	1	1
1	1	0		
1	1	1	1	1



4.6 Interpret a truth table

AQA EDUQAS OCR WJEC

isaacscs.org/bq/gcse_bool_05_v2

► Select the Boolean expression that corresponds to this truth table:

INPUTS		OUTPUT
A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

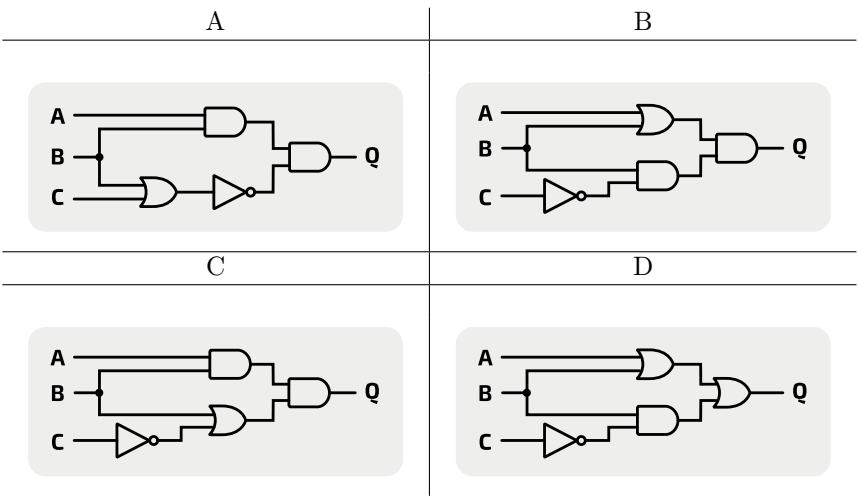
- ☐ $Q = \text{NOT } (A \text{ OR } B)$
- ☐ $Q = \text{NOT}(A \text{ AND } B)$
- ☐ $Q = A \text{ AND } (\text{NOT } B)$
- ☐ $Q = (\text{NOT } A) \text{ OR } B$

4.7 Wire this up!

AQA OCR

isaacscs.org/bq/gcse_bool_07

► Which of the circuits below represents the following Boolean expression?
 $=(A \text{ } B) (C+B)$



Answer:

4.8 Tables and gates

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_bool_09

A truth table is a table showing the outputs for all possible combinations of inputs to a logic gate or circuit. When putting values into a truth table, we often write them as 1 or 0. These values are interchangeable with True and False: 1 is True, and 0 is False.

Below we have an example of a logic gate:



► Circle the truth table that represent the logic gate provided:

Option A

A	B	Q
0	0	1
0	1	1
1	0	1
1	1	1

Option B

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

Option C

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

Option D

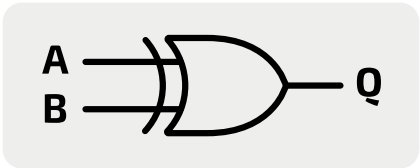
A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

4.9 The XOR gate (exclusive OR)

AQA EDUQAS WJEC

isaacscs.org/bq/gcse_bool_10

The following symbol represents an XOR gate (also known as Exclusive OR).



► Circle the correct truth table for an XOR gate from the options shown below.

Option A

A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

Option B

A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

Option C

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

Option D

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

4.10 Which Boolean expression?

AQA

isaacscs.org/bq/sys_bool_exp_truth_g1

► Which Boolean expression represents the function described by the truth table?

Inputs		Output
X	Y	Q
0	0	0
1	0	1
0	1	0
1	1	0

- ☐ $X \wedge \neg Y$
- ☐ $\neg(X \wedge Y)$
- ☐ $X \vee \neg Y$
- ☐ $\neg X \wedge Y$

4.11 Which Boolean expression 2

AQA

isaacacs.org/bq/sys_bool_exp_truth_g2

► Which Boolean expression represents the function described by the truth table?

Inputs		Output
X	Y	Q
0	0	0
1	0	1
0	1	1
1	1	0

- ☐ $\neg X \wedge Y$
- ☐ $X \vee \neg Y$
- ☐ $X \wedge Y$
- ☐ $X \vee Y$

4.12 Which logical operator?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/sys_bool_truth_g1

► Which logical operator has been used in the following truth table to produce output Q ?

Inputs		Output
X	Y	Q
0	0	0
1	0	1
0	1	1
1	1	0

- ☐ AND
- ☐ NOT
- ☐ XOR
- ☐ OR

4.13 Which logical operator 2?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/sys_bool_truth_g2

► Which logical operator has been used in the following truth table to produce output Q ?

Inputs		Output
X	Y	Q
0	0	0
1	0	0
0	1	0
1	1	1

- ☐ XOR
- ☐ AND
- ☐ NOT
- ☐ OR

5 Systems architecture

Systems architecture refers to the structure of the internal components of a computer system. Modern computers often have a **von Neumann architecture**, which involves: a processor; a memory unit, which stores both instructions and data; connections for input and output devices; and secondary storage for data. The **fetch-decode-execute cycle** describes how instructions are retrieved and processed in order to run a program. Several factors affect processor performance, including the number of cores and the size of the cache.

5.1 Hardware

AQA

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EDUQAS

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WJEC

isaacscs.org/bq/embedded_systems_g_01

► Embedded systems are often part of a larger computer system. From the options below, select 3 that describe the hardware components of embedded systems.

- ☐ All the components are on a single printed circuit board
- ☐ The components have very close control over the hardware
- ☐ They are specialised to the tasks they perform
- ☐ They have a high number of features and functions

5.2 Identifying embedded systems

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/embedded_systems_g_02

► Embedded systems can be hard to spot, and general-purpose systems often masquerade as embedded systems. From the list below, select 1 embedded system.

- ☐ A desktop computer
- ☐ Raspberry Pi
- ☐ A washing machine
- ☐ A laptop

5.3 Select the correct component

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_01

► Which one of the following CPU components is used to reduce how long the processor has to wait for data and instructions?

- ☐ Arithmetic and logic unit (ALU)
- ☐ Cache
- ☐ Control unit (CU)
- ☐ Bus

5.4 Decode stage processes

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_02

► Which one of the following statements describes the **decode** stage of the fetch-decode-execute cycle?

- ☐ The control unit (CU) works out what the instruction is about.
- ☐ The instruction to be executed is fetched from the main memory to the CPU.
- ☐ The results of the calculations that took place are stored back to the main memory.
- ☐ The arithmetic and logic unit (ALU) carries out the instruction.

5.5 Find the odd one out

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_06

► Which one of the below is **not** an example of an embedded system?

- ☐ Dishwasher
- ☐ Car navigation system
- ☐ Smartphone with a navigation app
- ☐ Car anti-lock braking system (ABS)

5.6 CPU performance

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_07

► Matthew is a web designer who uses processor-intensive applications (e.g. Adobe Photoshop). He wants to buy a new laptop and has the following options. Which one is the best choice for him?

- ☐ Clock speed = 3.5 GHz Cores = 6 Cache = 10 MB
- ☐ Clock speed = 3.6 GHz Cores = 2 Cache = 4 MB
- ☐ Clock speed = 4,600 MHz Cores = 4 Cache = 8 MB
- ☐ Clock speed = 1.2 GHz Cores = 4 Cache = 2 MB

5.7 A fetching problem

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_14

► During the fetch stage of the fetch-decode-execute cycle, a program instruction is fetched from the main memory. What is the name of the component that connects the processor to main memory?

- ☐ ALU
- ☐ Register
- ☐ Control unit
- ☐ Bus

5.8 Clock speed

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_15

► When comparing two processors which are identical except for the clock speed - which of the following statements will be true about the processor with the higher clock speed?

- ☐ Data needs to be fetched from RAM less often
- ☐ More instructions can be executed per second
- ☐ Programs take less time to run
- ☐ More cycles are completed per second

5.9 Tick-tock

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_sys_16

► The clock is used to synchronise the operations of the processor. Its speed is measured in cycles per second, and expressed in hertz (Hz), where 1Hz is one cycle per second. My computer system has a published processor speed of 3.3 GHz. How many cycles per second is my processor designed to operate at?

- ☐ 3,300
- ☐ 3,300,000,000
- ☐ 3,300,000
- ☐ 3.3

5.10 The decode stage

AQA

CIE

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_sys_17

► During the **decode** stage of the fetch-decode-execute cycle, which component of the CPU is used to send signals to other components on what needs to be done to execute the instruction?

- ☐ Main memory (RAM)
- ☐ Control unit (CU)
- ☐ Arithmetic and logic unit (ALU)
- ☐ Memory data register (MDR)

5.11 Processor performance

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/sys_arch_g01

► What may cause a **processor** to overheat?

- ☐ Reducing the amount of **cache memory**
- ☐ Increasing the number of **cores**
- ☐ Overclocking the **clock speed**

5.12 Prepare for processing

AQA

CIE

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_18

► The CPU is responsible for processing data and executing instructions inside a computer, there are a number of stages that instructions must go through before they reach the CPU. Which of the following statements about the instructions a CPU will execute are true? (Tick all that apply)

- ☐ The instructions are loaded into main memory from secondary storage
- ☐ The data required for the instructions has been loaded into main memory
- ☐ The instructions all need to be loaded into the processor at once so it can read them
- ☐ The user must enter their password to give permission for the instructions to be executed
- ☐ The instructions are translated into machine code that the processor can understand

5.13 Fetch, decode and execute

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_20

► Which of the following statements best describes the fetch-decode-execute cycle?

- ☐ The fetch-decode-execute cycle is a process that the CPU repeats continuously in order to execute instructions.
- ☐ The fetch-decode-execute cycle ensures improves the security of a computer system.
- ☐ The fetch-decode-execute cycle uses the the arithmetic logic unit (ALU) to connect the registers to the main memory.
- ☐ The fetch-decode-execute cycle uses a bus to store the data.

5.14 Storing instructions and data

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_19

► The term **Von Neumann architecture** is used to refer to any computer system where an operation to fetch an instruction and an operation to fetch data cannot occur at the same time because they share a common bus. Which of these statements about von Neumann architecture are correct? (Tick all that apply)

- ☐ Instructions and data are held in separate units of main memory.
- ☐ It uses registers to help prevent slowing down the fetch-decode-execute cycle.
- ☐ The program counter keeps track of the memory address of the instruction that needs to be executed next.
- ☐ The memory address register holds the address in main memory that is currently being read or written.
- ☐ The accumulator is a data register that can be used to store the results of calculations.
- ☐ The memory data register holds data fetched from memory or data waiting to be stored in memory.

5.15 Registers

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/sys_arch_g02

► Which sentence below best describes dedicated or special-purpose registers?

- ☐ Registers that are used to store the results of intermediate calculations that are part of larger computations.
- ☐ Processor-based registers that are used for a specific purpose.
- ☐ Temporarily holds the data (data values or instructions) that are read from or written to the main memory.
- ☐ Temporarily stores the memory addresses used when searching for data in RAM.

5.16 Getting what you pay for

AQA CIE EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_22

A civil engineer needs to use applications that require a high specification computer system (e.g. AutoCAD). She has only a basic laptop computer and its configuration is not good enough to run the applications, which are processor intensive. She has decided to buy a more powerful desktop computer and there are a few options available.

In general terms, the higher the clock speed, the more instructions a computer can process per second. However, other factors relating to the system’s architecture, such as cache and the number of cores, should be considered.

► Consider the options below and select the specification that will best match her needs to run processor-intensive applications.

☐ Option A

Computer specification	
Clock speed	3.6GHz

☐ Option B

Computer specification	
Clock speed	1.2GHz

☐ Option C

Computer specification	
Clock speed	4.6GHz

☐ Option D

Computer specification	
Clock speed	3.5GHz

5.17 Embedded systems

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/gcse_sys_21

An embedded system is a computer system dedicated to a specific task (or set of tasks). Embedded systems, just like general purpose computers, take input, run processes, and produce output. The embedded system combines the processor, memory, and input/output devices into a single unit.

Most modern day devices and appliances contain embedded systems, as they need some level of computational processing.

► Which of the options listed below would **not** be suitable for an embedded system?

- ☐ Central Heating
- ☐ Laptop
- ☐ Microwave
- ☐ Digital clock
- ☐ Medical Equipment
- ☐ Vending Machine

6 Memory and storage

Computers have two fundamental types of memory — main memory and secondary storage. **Main memory** is where data and instructions are stored so that they can be accessed directly by the CPU. **Secondary storage** is used to permanently store data such as the operating system and the user's files.

6.1 The purpose of RAM

AQA

CIE

EDUQAS

OCR

WJEC

isaaccs.org/bq/gcse_sys_11

► Select the statement that describes a purpose of Random-access memory (RAM).

- ☐ RAM is used to store the BIOS which is used when the computer is started.
- ☐ RAM is used to store instructions and data that the processor can directly access.
- ☐ CD-RWs are made up of RAM so that they can be rewritable.
- ☐ The hard disk drive (HDD) is made up of RAM so that data can be permanently stored.

6.2 Remember, remember

AQA

CIE

EDUQAS

OCR

WJEC

isaaccs.org/bq/gcse_sys_12

► Select four from the following statements about memory that are true:

- ☐ The contents of ROM are lost when power is switched off
- ☐ ROM is a type of primary memory
- ☐ RAM is volatile memory
- ☐ ROM can store the code needed to load the operating system into RAM
- ☐ Main memory is always volatile
- ☐ Virtual memory uses secondary storage to store data from the RAM

6.3 Back-up...

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_data_14

Marik is about to back-up his mobile phone data. The back-up storage app on his phone is reporting the following values in GB (gigabyte):

Description	Size
Photos	17.25 GB
Documents	3.25 GB
Messages	21.25 GB
Other data	2.75 GB

► Marik’s total internal back-up storage phone capacity is **50 GB**. How much free space is left of Marik’s back-up storage phone capacity?

- ☐ 5,500,000,000 bytes
- ☐ 5,000,000,000 bytes
- ☐ 5,000,500,000 bytes
- ☐ 44,500,000,000 bytes

6.4 Data storage units in order

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_data_18

► Arrange the below units of data storage into order from smallest to largest.

- ☐ 8000 MB
- ☐ 9 TB
- ☐ 7.8 GB
- ☐ 5.7 kB
- ☐ 2000 bits
- ☐ 200 bytes

6.5 How much space?

AQA

EDEXCEL

EDUQAS

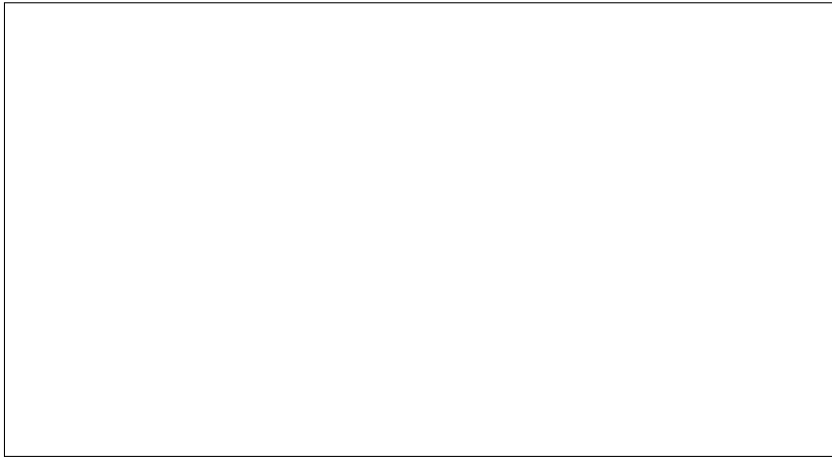
OCR

WJEC

isaacacs.org/bq/gcse_data_26

Tia is calculating the amount of disk storage that her bitmapped graphic will use. She has calculated that the total file size **in bits** as 12,200,000.

► How many **megabytes** of storage space will the file need?



6.6 Hard drive disadvantage

AQA

CIE

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_sys_05

► Which one of the statements below describes a disadvantage of magnetic hard drives (HDDs)?

- ☐ Hard drives are usually physically smaller than solid state drives, which means that they have less capacity for backing up data.
- ☐ Hard drives can be difficult to manufacture.
- ☐ It is significantly slower to access data stored on a hard drive than on a solid state disk because the hard drive has moving parts, such as the read/write head and the rotating platters.
- ☐ Hard drives are prone to scratches because they have a moving read/write head, and this can make parts of files unreadable.

6.7 RAM or ROM?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_sys_03

► Put the statements below into order to create one paragraph about the characteristics of **RAM** and then the characteristics of **ROM**.

- ☐ it is used to store software that doesn't need to be changed
- ☐ RAM
- ☐ the CPU is currently working with because
- ☐ such as the code needed to load the Operating System
- ☐ is a volatile type of memory and
- ☐ it is very useful for storing instructions and data
- ☐ when the computer is switched on.
- ☐ ROM
- ☐ is a non-volatile type of memory and
- ☐ it can be read from and written to.

6.8 Optical storage

AQA CIE EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_13

► Select three of the following statements about optical storage that are true:

- ☐ A CD-ROM is a type of optical media
- ☐ Optical media isn't easily portable due to the internal moving parts
- ☐ A laser beam burns tiny pits onto the disc
- ☐ Optical media can only be read-only memory
- ☐ Optical media has a limited amount of storage capacity compared to other types of storage
- ☐ Optical media is organised in groups of circular discs called platters

6.9 How many video files?

AQA CIE EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_08

► Pamela has a 128GB USB flash drive that is 60% full. How many video files can she store on the flash drive if each file is 25.6MB? Assume 1kB = 1000 bytes.

6.10 Which type of storage?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_sys_04

Laura has planned to go hiking in Snowdonia. She wants to take photos and videos of her trip with her digital camera and then show them to her friends, but her camera is running out of space.

► Which **type of storage** is the most suitable in this case?

► Which **storage device** would you recommend so that Laura does not run out of storage?

6.11 The levels of cache

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/sys_arch_memory_g_03

Cache memory is typically split into three levels. These levels have slightly different characteristics and enable the processor to save frequently used instructions.

► The levels have different access speeds. Number them in order from the fastest (1) to access to the slowest (3).

☐

Level 1

☐

Level 2

☐

Level 3

► The levels of cache also have different storage capacities. Number them in order from the largest (1) to the smallest (3).

☐

Level 1

☐

Level 2

☐

Level 3

6.12 What to store in RAM

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_arch_memory_g_01

► Random access memory (RAM) is used as “working memory” while a program is executed. Which parts of a computer program are held in RAM while it is processing?

- ☐ Intermediate values from calculations
- ☐ Instructions and data
- ☐ Only the instructions
- ☐ Only the data

6.13 Reasons for the cloud

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_arch_storage_g_04

► In recent years, the use of cloud storage has greatly increased. Which three of the following characteristics are important reasons to use cloud storage to store files?

- ☐ Sharing files
- ☐ Expandable storage capacity
- ☐ Safe back-ups
- ☐ Speed of access
- ☐ Cost

6.14 The largest memory

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_arch_memory_g_02

► There are many types of main memory, used for a variety of purposes. Which of the following is **most often** the highest capacity memory device inside a computer?

- ☐ Registers
- ☐ RAM
- ☐ Cache
- ☐ ROM

6.15 How to back up your archives

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/sys_arch_storage_g_01

Miguel works for a video production company and needs to clear up some space on his computer.

He has hundreds of gigabytes of video clips and projects that need to be backed up onto a storage device. The company does not want to spend too much money. They also would like to store the files in the office, to keep the files safe and easy to access. The new device does not need to be quick, as the files would first be transferred to a fast device installed on Miguel's computer before he uses them.

► Which three characteristics of a storage device are most important to Miguel and his company?

- ☐ Speed of access
- ☐ Portability
- ☐ Reliability
- ☐ Capacity
- ☐ Cost

► Which type of storage would you recommend for Miguel?

- ☐ Solid-state storage
- ☐ Magnetic storage
- ☐ Optical storage
- ☐ Cloud storage

6.16 Sharing a special moment

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/sys_arch_storage_g_02

Rosa has just graduated from university and her mother was there to capture it on video.

Now Rosa's family wants to see the video to celebrate with her. Many family members have asked Rosa to send them a copy of the video so they can share in her triumphant moment. However, Rosa has lots of relatives and not all of them have access to an internet connection. They all have access to storage connections such as a USB and CD drive. The video is only a few megabytes in size.

► Which three characteristics are most important to Rosa for selecting a storage device?

- ☐ Speed of access
- ☐ Portability
- ☐ Reliability
- ☐ Cost
- ☐ Capacity

► Which type of storage would you recommend for Rosa?

- ☐ Optical storage
- ☐ Magnetic storage
- ☐ Cloud storage
- ☐ Solid-state storage

6.17 Make it faster

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_arch_storage_g_03

Mads works as a product designer for a fashion retail company.

He frequently uses an application to create 3D designs for clients, but he is having issues with it. The application is slow to load and often takes minutes to save his progress. One day after losing a file for the fourth time, Mads decides he needs a new storage device to save his application on.

His company has offered to pay for a new storage device if it will help Mads do his work, so money is no concern. The program is quite large and complex.

► Which three characteristics are most important to Mads for selecting a storage device?

- ☐ Speed of access
- ☐ Capacity
- ☐ Cost
- ☐ Portability
- ☐ Reliability

► Which type of storage would you recommend for Mads?

- ☐ Optical storage
- ☐ Solid-state storage
- ☐ Cloud storage
- ☐ Magnetic storage

7 Hardware

Hardware is the word used for the physical components of a computer system. Hardware can refer to the **internal** parts of the computer (motherboard, CPU, and RAM), or **external** devices, known collectively as peripherals. **Peripheral devices** can be used to input data (keyboards, microphones, scanners), output data (monitors, printers, speakers), or to store data (external hard drives, USB memory sticks).

7.1 PCBs

EDUQAS WJEC

isaacscs.org/bq/sys_hard_g01

► Which of the following components would be considered to be a printed circuit board (PCB)? Select **one**.

- ☐ CPU
- ☐ Graphics processing unit
- ☐ Fan
- ☐ Graphics card

7.2 Input devices

EDUQAS WJEC

isaacscs.org/bq/sys_hard_g02

► Which of the following input devices would be most suitable to open an app on a smartphone? Select **one**.

- ☐ Keyboard
- ☐ Mouse
- ☐ Trackpad
- ☐ Touchscreen

7.3 Selecting a printer

EDUQAS

WJEC

isaacss.org/bq/sys_hard_g03

► Aanya has recently set up her business as a professional photographer. She needs a printer that will print her photographs in the best possible quality. Which type of printer is the most suitable for Aanya? Select **one**.

- ☐ 3D printer
- ☐ Inkjet printer
- ☐ Plotter
- ☐ Laser printer

7.4 Input devices

EDUQAS

WJEC

isaacss.org/bq/sys_hard_g04

► Which **two** of the following statements are **true**?

- ☐ A mouse and a microphone are both input devices.
- ☐ A mouse and a monitor are both input devices.
- ☐ A touchscreen monitor can be both an input and output device.
- ☐ A mouse and a printer are both input devices.

7.5 Mother, I'm bored!

EDUQAS

WJEC

isaacss.org/bq/sys_hard_g10

► Which component on a motherboard carries data?

- ☐ RAM
- ☐ Bus
- ☐ CPU

7.6 Output devices

EDUQAS WJEC

isaacacs.org/bq/sys_hard_g05

► Put the devices into the correct position in the table to match their purpose.

Purpose	Device
Uses one or more pens to draw an image. Pens are lifted or applied to the paper as required to produce very precise drawings.	
Generates sound waves from data being received from a sound card. Often built into a computer system, but can also be connected as external peripherals.	
Allows for three-dimensional designs on a computer to be printed in a solid form. The device uses filament, usually in the form of plastic-based resin, which is heated and printed layer by layer to form a solid object.	
Prints a whole page at a time and is typically capable of printing hundreds of pages per minute. This device uses powdered ink called toner.	
Uses liquid ink to produce black-and-white or coloured prints. Liquid ink produces rich colours, meaning it is the most suitable for printing photographs.	

- ☐ Laser printer
- ☐ Plotter
- ☐ 3D printer
- ☐ Speaker
- ☐ Inkjet printer

7.7 RFID tags

EDEXCEL

isaaccs.org/bq/sys_hard_g06

RFID tags are made up of an antenna — for receiving and transmitting the radio frequency signals — and a chip, which processes the signals. Each tag has a unique identifier and will often store some additional data in a non-volatile memory cell.

► Read through the statements below and put the words provided into the correct places to fill the gaps.

Passive devices do not contain a _____. Instead, the device is powered by _____ transmitted by the reader. The antenna collects the energy from the transmitter to activate the _____.

Active systems have a small battery within the tag, and the device will transmit its _____ at regular intervals. Active RFID devices are commonplace in shops, where _____ tags are secured onto items. The tags are deactivated or removed when the goods are paid for. A _____ is situated at the shop's exit and an alarm is triggered if it detects an activated tag when a customer leaves the shop.

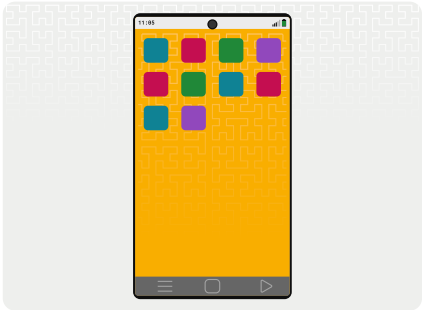
- ☐ power supply
- ☐ radio energy
- ☐ chip
- ☐ identifier
- ☐ battery-powered
- ☐ reader

7.8 So you think you are smart?

EDUQASWJEC

isaaccs.org/bq/sys_hard_g08

Below is an example of a smartphone:



► Using the table below, state whether each of these components of a smart-phone is an **input**, **output**, or **both**:

Feature	Input/Output/Both
Speaker	
Microphone	
Touchscreen	
Camera	

7.9 Accessible input

EDUQASWJEC

isaaccs.org/bq/sys_hard_g09

► Bryn has locked-in syndrome, a severe motor disability that means he has no fine motor control and cannot use conventional input devices. Which **hardware** device listed below would be the most appropriate in enabling Bryn to interact with and use his computer?

- ☐ Sip/puff switch
- ☐ Keyboard
- ☐ Touchscreen

7.10 In, out, shake it all about

EDUQAS WJEC

isaaccs.org/bq/sys_hard_g07

► Complete the table below by labelling each device as either input or output.

Device	Input or Output
Monitor	
Printer	
Keyboard	
Mic	
Speaker	

8 Software

Software is the word used to describe the programs that provide the instructions to control the computer hardware components. There are several broad categories of software: **application software** provides a tool for a specific purpose (for example, a word processor), **system software** controls the hardware (for example, an operating system) and **utility software** helps configure and maintain the computer.

8.1 Categories of software

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaaccs.org/bq/sys_os_software_defs_g01

► Which of the following categories of software would an operating system be classed as?

- ☐ Utility software
- ☐ System software
- ☐ Application software

8.2 Disk defragmentation

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaaccs.org/bq/sys_os_software_utility_g01

► What processes does disk defragmentation follow? Number the statements in the correct order.

- ☐ Files are temporarily moved to a "scratch" part of the hard drive
- ☐ Related data is collected together
- ☐ The empty spaces are collected together
- ☐ Data is replaced on the main section of the hard disk

8.3 Apps for who?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_soft_g01

► Which one of these uses application software to perform tasks?

- ☐ Operating system
- ☐ System software
- ☐ Utilities
- ☐ Users

8.4 Sending files

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_soft_g02

► Fill in the gap using one of the words provided.

The time required for users to download a file can be reduced by using _____ software on the file before it is uploaded.

- ☐ encryption
- ☐ compression
- ☐ defragmentation

8.5 Dealing with corrupt files

EDEXCEL

EDUQAS

WJEC

isaacacs.org/bq/sys_soft_g05

► Which one of these utilities should be run on a corrupt file?

- ☐ Compression
- ☐ Defragmentation
- ☐ File repair
- ☐ Encryption

8.6 Defrag all the things?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/sys_soft_g04

► Which **one** of these is a reason why the performance of a solid-state drive is not improved by running a defragmentation utility on it.

- ☐ Files stored on a solid-state drive cannot be fragmented
- ☐ Solid-state drives already have faster read times than magnetic drives
- ☐ Utility software can only be used with magnetic media
- ☐ Solid-state drives have no moving parts

8.7 Locked down

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/sys_soft_g06

► Complete this sentence by adding the missing word into the space provided.

An encryption algorithm, also called a _____ is used to encrypt plain text.

8.8 In case of emergency!

AQA

EDEXCEL

EDUQAS

WJEC

isaaccs.org/bq/sys_soft_g07

► Complete this sentence by adding the missing word into the space provided.

Computers use _____ utilities so that data can be restored in the case of a system failure.

8.9 Is your knowledge secure?

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/sys_soft_g08

► Which **one** of these is used by an anti-malware utility?

- ☐ Encryption software
- ☐ System cleanup tools
- ☐ A directory of known viruses
- ☐ Corrupt data

8.10 Order! Order!

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/sys_soft_g03

Two files (A and B) are stored on a hard disk.

Each file is split into two blocks.

► Rearrange the file blocks to show the arrangement after a defragmentation utility runs.

--	--	--	--

9 Operating systems

The **operating system** is the most fundamental piece of **system software**; it manages interactions between software and hardware. It takes responsibility for a range of functions including managing processor time, memory management, inputs/outputs, and security.

There are lots of different operating systems. Some are custom written for a particular device, such as a games machine or the guidance system on the International Space Station. Other operating systems are used across a range of devices. The most popular and well-known operating systems are; Microsoft Windows, Apple Mac OS, Linux and Google's Android.

9.1 Roles of the operating system

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/sys_os_role_of_os_g01

► Which of the following tasks is **not** one of the roles of the operating system?

- ☐ Malware protection
- ☐ Controlling input and output devices
- ☐ Memory management
- ☐ Processor management

9.2 Multi-tasking

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/sys_os_role_of_os_g02

► Which of the following terms best describes **multi-tasking**?

- ☐ Tasks being swapped in and out of the processor so that it appears to the user that the computer is carrying out more than one task at the same time.
- ☐ The process of interrupting a task being carried out by the processor so that a new task with a higher priority can be completed.
- ☐ Multiple core processors carrying out tasks simultaneously
- ☐ Tasks being swapped in and out of memory simultaneously

9.3 Device Drivers

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_os_role_of_os_g03

- Which of the following best describes the purpose of a **device driver**?
- ☐ Hardware built into devices such as motor to enable movement
 - ☐ Software translates commands from the peripheral device into a form that the operating system and other programs can understand
 - ☐ Hardware that receives commands from the operating system
 - ☐ Software included with office based applications to allow printing.

9.4 Peripheral software

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_os_g06

An operating system (OS) is software that controls the computer's hardware and software resources. One role of the operating system is to communicate with and manage peripherals such as printers.

- Identify the correct piece of software that allows the operating system to interact with peripherals.
- ☐ Device driver
 - ☐ Firewall
 - ☐ Disk defragmenter
 - ☐ Memory manager

9.5 All those tasks!

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_os_g07

- Which **four** of the tasks below are carried out by the operating system?
- ☐ Transfers data to a printer
 - ☐ Creates new folders
 - ☐ Provides an interface for the user to interact with the computer
 - ☐ Sets automatic backups for data
 - ☐ Uses a scheduler to swap tasks in and out of the processor
 - ☐ Deletes emails

9.6 Operating system actions

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/sys_os_g05

The table below identifies actions of an operating system. Each action is managed by a **different** function of an operating system: memory, processor, or file.

► Complete the table by adding a **tick** to identify what actions are managed by each function and a **cross** to identify what actions are not managed by each function.

Action	Memory management	Processor management	File management
Delete a file			
Load data from secondary storage into the main memory			
Determine which portion of memory should be available for a process			

9.7 Access levels

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/sys_os_g01

A student is considering training to become a teacher, so has offered to volunteer at a local primary school to gain some experience.

They need access to the school’s computer system in order to see lesson resources.

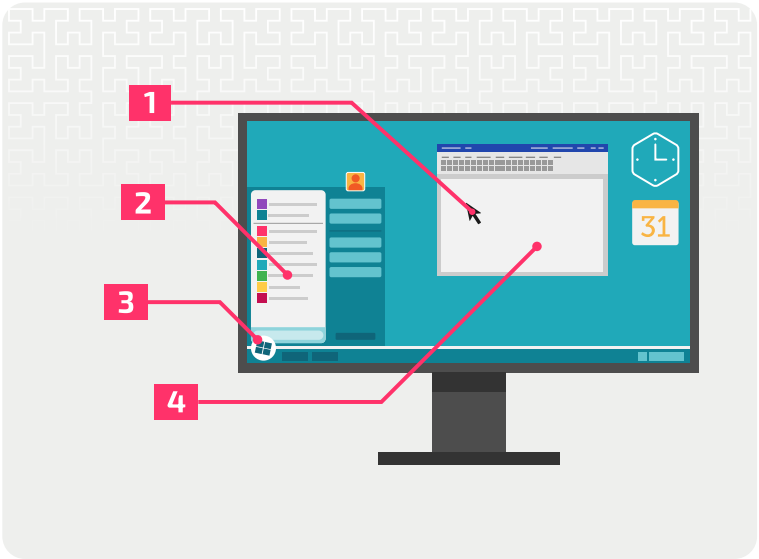
- What level of access would be appropriate for the school to give the student?
- ☐ Write
- ☐ Modify
- ☐ Read

9.8 Gooney GUI!

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/sys_os_g02

► Below is an image of a graphical user interface (GUI) with some features numbered.



Add the correct label into the corresponding position in the table:

Number	Correct label
1	
2	
3	
4	

☐ Windows

☐ Menus

☐ Pointer

☐ Icons

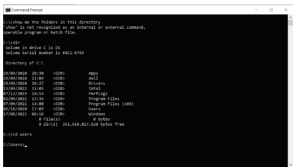
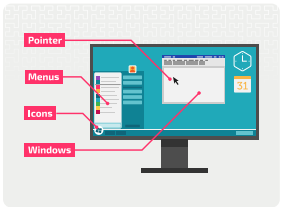
9.9 What interface?

- AQA
- EDEXCEL
- EDUQAS
- OCR
- WJEC

isaaccs.org/bq/sys_os_g04

To allow the user to interact with the computer, an operating system provides a user interface (UI). There are various forms that a UI can take.

► Correctly match the UI to the image that represents it using the following options; **Command line interface (CLI)**, **Natural language interface** and **Graphical user interface (GUI)**.



9.10 Stop interrupting me!

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/sys_os_g03

► The table below contains definitions of some basic functions of an operating system. Complete the table by matching the definition to the function.

Function	Definition
	Determines how much memory to allocate to each process.
	Allocates time to processes that are deemed urgent.
	Swaps tasks in and out of the processor so it appears the computer is doing more than one task at the same time.

- ☐ Interrupt
- ☐ Memory manager
- ☐ Scheduler

10 High- and low-level languages

Most programming languages that you will have heard of are high-level languages. Python and C# are examples of high-level languages that are widely used in education and in the workplace. A **high-level language** is one that is user-oriented in that it has been designed to make it straightforward for a programmer to convert an algorithm into program code.

A **low-level language** is machine-oriented. Low-level programs are expressed in terms of the machine operations that must be performed to carry out a task. This makes writing programs more difficult, as the algorithm must be specified in terms of the capabilities and specifications of the processor. Low-level languages are named for the processor (or processor family) that they are designed for, and are often referred to as **assembly language** or **machine code**.

10.1 High-level characteristics

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_proglang_g01

► Which **three** of the following are characteristics of a high-level programming language?

- ☐ Uses an assembler to convert to machine code
- ☐ Uses syntax which is similar to human language
- ☐ Instructions are written in assembly language
- ☐ Hardware is independent so can be used on multiple devices
- ☐ Instructions are written in source code

10.2 Difficult programming languages

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_proglang_g06

► Which type of programming language is widely considered to be **easier**, because of its use of common language commands and built-in libraries?

- ☐ High-level language
- ☐ Low-level language

10.3 High- or low-level?

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/sys_proglang_g02

► Label the statements below showing the characteristics of low- and high-level programming languages by adding **High level** or **Low level** into the last column.

Characteristic	Label
Use syntax similar to human language	
Require specific knowledge of a processor and its operations	
Must be compiled and converted into machine code before they are run	
Typically used to control specific hardware	
Have built-in libraries that the programmer can call upon	
Are easier to learn and understand	
Sometime use a set of mnemonics for key commands	

10.4 Why high-level?

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/sys_proglang_g03

► Blake is developing an app for their Computer Science school project and the teacher has suggested they use a high-level language to do this. What would be the **three** best reasons for this suggestion?

- ☐ It is written in assembly language which Blake has recently learnt
- ☐ It will include libraries that may be useful in developing the app
- ☐ It will be easier to code and develop as a relative beginner
- ☐ It is high-level so is complicated and will demonstrate their coding capability
- ☐ It will be portable and could be used on multiple devices

10.5 Identifying languages

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/sys_proglang_g04

► Below are some examples of programming languages. Identify the correct classification (high- or low-level) to match each example.

```
first = input()
second = input()
result = first + second
print(result)
```

```
INP
STA 99
INP
ADD 99
OUT
HLT
```

```
10010000001
00111100011
10010000001
00011100011
10010000010
00000000000
```

10.6 Which language to use?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacCS.org/bq/sys_proglang_g05

► Below are some examples of programming tasks. Add the correct type of language to use for the task (high- or low-level) into the table.

Example	Classification
Creating a mobile app to track the amount of water people are drinking	
Writing a program for a embedded system that measures the moisture in a greenhouse	
Creating a device driver for your custom keyboard	
Creating a 3D video game for your friends	
Creating a motion-controlled light with a microcontroller	

10.7 Compiler vs. interpreter

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacCS.org/bq/sys_proglang_g07

► When using high-level languages, it is necessary to convert code into machine language before it can be run. This can be done by either a **compiler** or **interpreter**. Look at the statements below and decide which is the correct definition for Compiler and Interpreter.

Converts the whole code into machine code before running it

Converts one statement of the program at a time into machine code

10.8 Low-level characteristics

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/sys_proglang_g08

► Which **three** of the following are characteristics of a low-level programming language?

- ☐ It has a 1:1 relationship with processor instructions
- ☐ Allows for individual control of a processor’s components and registers
- ☐ Instructions are written in assembly language or machine code
- ☐ Uses syntax that is similar to human language
- ☐ Hardware is independent so can be used on multiple devices

10.9 Is it high or low?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/sys_proglang_g09

Programming languages are considered to be either high level or low level. Using the names of the programming languages provided, add the language into the correct position in the table to state whether it is high or low level.

High level	Low level

- ☐ Machine code
- ☐ C Sharp
- ☐ Assembly language
- ☐ Python

10.10 Assembly language facts

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/sys_proglang_g10

► Below are some statements about assembly language. Use the words provided to complete the sentences.

Assembly languages allow programmers to write _____ using a set of _____ that represent the _____ equivalent.

In assembly language, the mnemonics are an _____ language representation of the operation the machine will perform when executed.

Assembly language needs to be converted into machine code for it to run. This is done by _____ the source code into the machine code equivalent of the assembly language program.

This process is called _____ and is done by a program called an assembler.

- ☐ binary
- ☐ English
- ☐ machine code
- ☐ assembling
- ☐ translating
- ☐ mnemonics

11 Translators

A **translator** is a piece of software that handles the translation of source code into a form that can be executed by a processor. **Interpreters**, **compilers**, and **assemblers** are all types of translator. Depending on the situation, you will need to use different types of translators. You will also explore the multiple stages of translation between source code and object code.

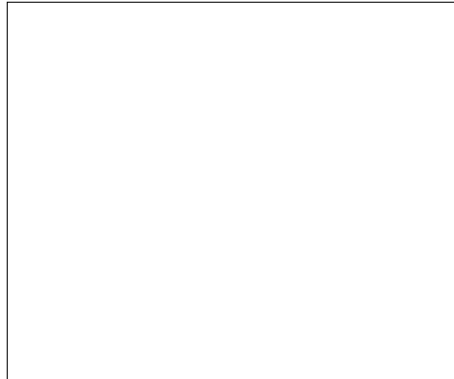
11.1 Stages of compilation

EDUQAS WJEC

isaacacs.org/bq/sys_trans_g01

► Put the following stages of compilation into the correct order.

- ☐ Semantic analysis
- ☐ Syntax analysis
- ☐ Code optimisation
- ☐ Create source code
- ☐ Code generation
- ☐ Lexical analysis



11.2 Which translator type?

EDUQAS WJEC

isaacacs.org/bq/sys_trans_g02

► Select the type of translator represented by this description: "Translates code into a form that **can** be executed. However, it does not execute the code."

- ☐ Compiler
- ☐ Interpreter
- ☐ Assembler

11.3 Describing an interpreter

EDUQAS WJEC

isaacccs.org/bq/sys_trans_g03

- Select the correct statement that describes an interpreter.
- ☐ The translation process is a relatively simple one with this translator. There is a one-to-one relationship between the code and the machine-executable code.
 - ☐ This type of translator produces executable code in a form that cannot be read by a human or readily reverse-engineered (i.e. converted back into source code).
 - ☐ Following the syntax check, each line of source code is converted into its machine code equivalent and executed. If a runtime error occurs, the program crashes.

11.4 Translator drawbacks

AQA EDEXCEL EDUQAS OCR WJEC

isaacccs.org/bq/sys_trans_g04

- Select the **two** statements that describe drawbacks of using a **compiler**.
- ☐ You have to wait until the entire program is translated before receiving any errors.
 - ☐ If you want to change something in the program, you have to translate the whole program again.
 - ☐ The source code must be given to the user.
 - ☐ The source code is translated line by line while execution is underway, so the program can run more slowly.

11.5 Which stage?

EDUQAS WJEC

isaacccs.org/bq/sys_trans_g06

- Which stage of the compilation process ensures that the tokens follow the rules of the programming language being used?
- ☐ Lexical analysis
 - ☐ Syntax analysis
 - ☐ Semantic analysis

11.6 Compiler or interpreter?

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/sys_trans_g05

High-level languages need to be translated. This can be done by either a **compiler** or an **interpreter**.

- Select the **three** statements that are features of a compiler.
 - ☐ Any errors are reported at the end of the process.
 - ☐ All code is translated prior to being executed.
 - ☐ Only an executable file is given to the user, not the original source code.
 - ☐ Errors are discovered line by line, so can be corrected as they are detected.
 - ☐ Code is translated line by line while execution is underway.
 - ☐ The source code must be given to the user.

11.7 Compiler vs. interpreter

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/sys_trans_g07

► When using high-level languages it is necessary to convert code into machine language before it can be run. This can be done by either a **compiler** or **interpreter**. Match the definitions below to either Compiler or Interpreter.

Converts the whole code into machine code before running it

Coverts one statement of the program at a time into machine code

12 Social engineering

Social engineering is the term used for a range of techniques employed by cybercriminals to deceive users into giving away their IDs, passwords, or other personal information. Criminals who are intent on getting access to our data come up with new ways of tricking us every day. Techniques that you may have heard of include **phishing**, **shouldering**, **blagging**, and **pharming**.

Social engineering is different to other cybercrimes because it involves humans trying to **trick** or **manipulate** other humans.

Technological solutions can be put in place to prevent unauthorised access to data, but it is arguably **human users** that pose the most significant risk to a computer system. For example, validation rules can be put in place to specify that a password must be a certain length and include special symbols and numbers, however, no IT system can prevent a user from accidentally revealing their username and password to another person.

Fortunately, there are ways to avoid being caught out. Much of this relies on general awareness, but there are also software solutions that can detect and alert users to potential scams.

12.1 Banking fraud

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/cyb_social_techniques_g03

An online bank has the URL **<https://oceansavings.org>**.

One of their customers has recently had their login details stolen as they accidentally entered their username and password into a fake website that looked identical to the website of the real bank. By accident, they had entered the following URL **<https://oceansavings.com>**.

► What is the name for this type of **social engineering** technique?

- ☐ Phishing
- ☐ Pharming
- ☐ Blagging
- ☐ DDoS attack

12.2 Gone phishing

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/cyb_social_techniques_g01

- From the list provided, select the **two** indicators of a **phishing** email:
- ☐ Email sent at unexpected time of day (i.e. outside of the normal working day)
 - ☐ Email encourages the recipient to click on a suspicious looking hyper-link
 - ☐ Email attempts to initiate a conversation with the recipient
 - ☐ Email does not address recipient by name

12.3 Social engineering attacks

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/cyb_social_techniques_g02

- The table below provides definitions for common social engineering techniques. Choose from the words below to match the technique to the definition.
- ☐ Phishing
 - ☐ Blagging
 - ☐ Pharming
 - ☐ Shouldering

Technique	Definition
	An attack in which the victim receives a message disguised to look like it has come from a reputable source (for example, a bank), in order to trick them into giving up personal information.
	An attack in which the perpetrator invents a scenario in order to convince the victim to give them personal information or money.
	Deceiving users by sending them to a fake website that the user believes is the real one, with the intention of tricking them to submit personal data.

12.4 A flight to Dubai

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/cyb_social_techniques_g05

Michael is a successful engineer and works for a multinational company. His laptop is full of sensitive information and confidential business operations.

During a business trip to Dubai, Michael noticed that the passenger next to him was staring at his laptop screen as he started to enter his username and password. Michael stopped typing his details and closed his computer.

► Based on the given scenario, select which type of cyberattack Michael could have been a potential victim of?

- ☐ Shouldering
- ☐ Blagging
- ☐ Pharming
- ☐ Phishing

12.5 Open your eyes!

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/cyb_social_techniques_g06

Angela has joined a gardening group on a social media platform. A group member asked Angela for her email and mobile number, so Angela passed her details to her new friend, and conversations took place for over a month on a daily basis.

One day the friend told Angela that they had been in a car accident and needed to borrow money from her to pay for the hospital bill. Angela has a big heart and quickly sent the money to a bank account provided by her friend.

Angela didn't receive anymore messages from her friend and did not get any money back.

► Based on the given scenario, which type of cyberattack did Angela suffer?

- ☐ Phishing
- ☐ Pharming
- ☐ Shouldering
- ☐ Blagging

12.6 A taste of their own medicine

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/cyb_social_techniques_g07

The Department of Defence decided to use cyberattack techniques against hackers. The cybersecurity team created an advertisement where the perpetrators supposedly could be hired to perform cyberattacks.

By responding to the advertisement, the hackers had a Trojan Horse installed on their computers. This virus/trojan then changed the computer's host file to direct traffic away from its intended target, and toward a fake website instead. This operation was very successful, and many criminals were caught and brought to justice.

► Based on the given scenario, which type of cyberattack did the Department of Defence use to catch the hackers?

- ☐ Blagging
- ☐ Shouldering
- ☐ Pharming
- ☐ Phishing

12.7 It's too good to be true!

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/cyb_social_techniques_g08

Martha has received an email that appears to be from Amazon saying that she has won a £500 Amazon gift card. To claim the voucher, she needs to fill out a form with her name, address, date of birth, and Amazon account username.

Martha is not good with computers but she does think this email could be too good to be true. She shows the email to her daughter who advises her to delete the email and block the email account.

► Based on the given scenario, which type of social engineering attack has Martha been a potential victim of?

- ☐ Shouldering
- ☐ Pharming
- ☐ Phishing
- ☐ Blagging

12.8 Security techniques

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/cyb_social_techniques_g04

► Add the correct terms to complete the text below:

_____ is a term used to describe a broad range of malicious activities accomplished through human interactions. These techniques are designed to deceive users into giving away personal information, passwords, or user IDs.

In some cases, it is possible to prevent these attacks by taking some security measures and using strong _____ techniques.

Users need to be vigilant and aware of how to spot these forms of attack. _____ emails, for example, often have poor grammar and spelling, and may not be addressed directly to the recipient.

- ☐ authentication
- ☐ Social engineering
- ☐ Phishing

13 Malicious software

Malicious software (malware) is created with the intent to cause damage to computer systems or steal data, and computer users must take steps to lessen these threats. There are many different forms of malware, some of which can be classified as **viruses**, **trojans**, and **worms** based on how they behave. These are explained further in the sections below.

Different forms of malware have different purposes. **Spyware**, for example, can be used to track your use of computer systems, and **keyloggers** can capture and transmit any data that you enter into a computer system. **Ransomware**, by contrast, prevents a user accessing their computer files.

Fortunately, for every threat, there is a form of defence. It is essential that everyone who uses a computer recognises how their system can be exposed to threats, and understands the importance of keeping software up to date and regularly scanning for malware.

13.1 Which attack?

AQA EDEXCEL EDUQAS OCR WJEC

isaacss.org/bq/cyb_mal_g01

► Using the statements provided, **match** the form of attack to the correct description.

Form of attack	Description
	Replicates and is designed to infect as many systems as possible. Does not require a host program.
	Appears to be a legitimate file, but actually performs malicious actions.
	Replicates and attaches to other programs or files to spread over computer systems.

- ☐ Worm
- ☐ Virus
- ☐ Trojan

13.2 Features of malware

AQA EDEXCEL EDUQAS OCR WJEC

isaacss.org/bq/cyb_mal_g02

► It is important to know the difference between the types of malicious software and how they operate. Read the statements below and then fill in the gaps using the words provided.

Malware is designed to gain access to a computer system with _____ intent. Viruses and worms are both _____, which means that they can spread quickly across a computer system.

Unlike a _____, a _____ requires a host file, and will activate when the program is run. A _____ is a form of malware that appears to be a legitimate file, but unbeknown to the user, it also performs malicious actions such as data theft.

Spyware is a type of malicious software that leaks data back to the creator, a _____ is a type of spyware that records keys pressed to discover passwords and confidential information.

- | | |
|------------------------------------|---|
| ☐ trojan | ☐ keylogger |
| ☐ worm | ☐ self-replicating |
| ☐ malicious | ☐ virus |

13.3 Someone call security!

AQA EDEXCEL EDUQAS WJEC

isaacss.org/bq/cyb_mal_g03

► Identify **two** ways in which security flaws in software can be identified and removed to prevent the impact of malware.

- ☐ Acceptable use policies
- ☐ Anti-malware software
- ☐ Patching
- ☐ Code review

13.4 Payment due!

AQA

EDEXCEL

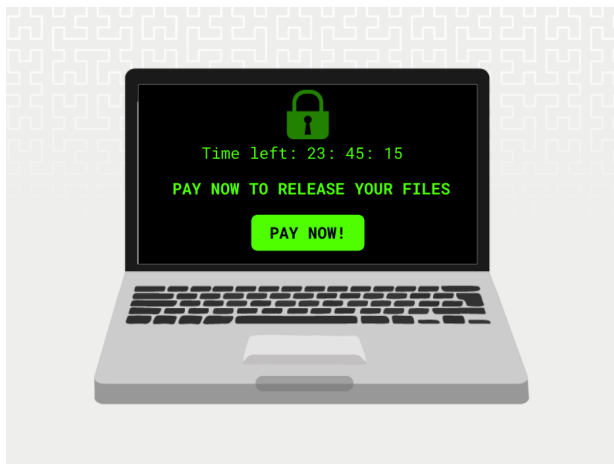
EDUQAS

OCR

WJEC

isaacss.org/bq/cyb_mal_g04

Farah was using her laptop to complete some online research for her home-work when the following message popped up on the screen:



► What form of malware attack is this?

- ☐ Virus
- ☐ Ransomware
- ☐ Spyware
- ☐ Trojan

► What would be the **two** best methods to prevent this type of attack and its impact?

- ☐ Regular backups
- ☐ Security patches
- ☐ Acceptable use policies
- ☐ Anti-malware software

14 Network security

Computer networks are a part of everyday life, and many of us connect to some kind of network on a regular basis. You may keep much of your data “in the cloud” or use online banking to manage your money. Several online stores may have a record of your card payment details.

Networks contain so much valuable data and, unfortunately, there are many criminals who would love to get their hands on it. If the network is connected to the internet, criminals can hack accounts and launch cyberattacks from anywhere in the world. This means that they can be hard to track down and hard to prosecute.

Access to networks can be controlled by ensuring that users have strong passwords, and by setting appropriate access rights for authorised users. Codes of conduct and acceptable use policies can be set to ensure that users follow guidelines for keeping corporate data safe, and to hold them accountable if they don't. Backup systems and recovery procedures ensure that essential data can be recovered if the worst happens.

14.1 Strong passwords

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/cyb_sec_g01

It is good practice to use strong passwords, i.e. ones that are difficult to guess. A long password is usually strong because it would take more attempts to guess the password. However, other factors are also important.

► Disregarding length, which of the following passwords is the strongest?

- ☐ I*saacC£S9
- ☐ The name of your favourite football team combined with your date of birth
- ☐ zxcvbnm
- ☐ flibbertigibbet123

14.2 Avoiding the queues

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/cyb_sec_g03

► Airports and airlines worldwide are investing in alternative means of suitably identifying and checking the security status of passengers. Often, the procedure consists of getting information validated in advance, which enables passengers to pass through a checking point without ever showing their passport or ticket. Which type of authentication can be used in this case? Select the correct option.

- ☐ Password protection
- ☐ Two-factor authentication (2FA)
- ☐ Biometric authentication
- ☐ Physical security

14.3 Filtering it

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/cyb_sec_g04

► Fill in the concept that correctly matches the description provided in the table below.

Description	Concept
Term used to describe the MAC addresses with special rights or privileges.	
Term used to specify the MAC addresses of devices that are allowed to connect to the network.	
Term used to specify the MAC addresses of devices that are not allowed to connect to the network.	
Security solution, although not very efficient, as it is possible for a device to spoof an authorised MAC address and gain access to the network unnoticed.	

- ☐ Blocklist
- ☐ Safelist
- ☐ Whitelist
- ☐ MAC filtering

14.4 Symmetric vs asymmetric

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/cyb_sec_g05

► Complete the table below by using the numbered statements to show the differences between the symmetric and asymmetric encryption techniques.

Differentiator	Symmetric encryption	Asymmetric encryption
Number of keys		
Speed of execution		
Security		
Length of keys		

- 1. The secret key is shared, so the risk of compromising is higher
- 2. The length of the keys used is larger, typically 2048 bits or higher
- 3. Faster because it uses simple mathematical operations
- 4. Uses the same key for both encryption and decryption
- 5. Slower, but allows the sender to be authenticated
- 6. The length of the keys used is typically 128 or 256 bits
- 7. The private key is not shared, and the overall process is more secure
- 8. Uses a related set of keys, one public and one private

14.5 Working from the coffee shop

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/cyb_sec_g07

The company ‘Security Guru’ recently installed a Wi-Fi system to replace their old wired network. The new network uses radio frequencies to transmit data, and offers greater flexibility and a reduction of costs. James works as a financial adviser at Security Guru and is required to log in to use the company’s Wi-Fi, called **security_guru**, to connect to the internet. The network is encrypted end-to-end and the Wi-Fi uses WPA3 technology, which provides strong protection, such as individual traffic encryption and protection against brute force attacks.

During his break, James crossed the street and went to the coffee shop to have something to eat. As he was waiting for his order, he realised that he had forgotten to send an important email to the accountancy department with the company’s monthly financial balance. James tried to connect to the internet using his mobile phone, but the company’s Wi-Fi **security_guru** was out of range. However, the network **secur1ty_guru_guests** was openly available, not requiring any type of authentication. So, James connected to the network, finished writing the email with the financial balance sheet attached, and sent it to the accountant’s office.

► Based on the scenario above, select all the options that represent what James should or should not do to keep the company network safe from cyberattacks.

- ☐ James could connect to the network ‘secur1ty_guru_guests’ without concerns as it is very secure as the new Wi-Fi system offers strong protection against attacks.
- ☐ James could connect to the network ‘secur1ty_guru_guests’, but only to check his personal email and to log into his social media accounts.
- ☐ James could connect to the network ‘secur1ty_guru_guests’, but he needs to know that it does not protect against attacks.
- ☐ James should not connect to the network ‘secur1ty_guru_guests’ to send his business email or any important or sensitive information.

14.6 Testing from inside out

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/cyb_sec_g08

The following describes some examples of how criminals explore human and technical vulnerabilities to gain unauthorised access to data.

Human vulnerability

- Scanning social media for hobbies, interests, and home address
- Scanning for personal and work email addresses to identify friends and colleagues
- Scanning for personal and work phone numbers to help identify linked accounts and associates
- Scanning online activity to find usernames and passwords
- Scanning for work colleagues, their responsibilities, and hierarchy

Technical vulnerability

- Scanning the system to identify weak spots and open doors
- Gaining access to the system then expanding access once within it
- Extracting copies of data, documents, and/or information for future analysis and exploitation
- Establishing a route to get back into the system easily in future should this be required
- Covering tracks, so the means of access and activities carried out are hard to determine even for an expert

► Which type of penetration test can be done to simulate or identify cyber-attacks from an internal perspective, where the attacker already has knowledge about and access to the system and knows how the network topology is designed?

- ☐ Black-box penetration testing
- ☐ White-box penetration testing
- ☐ Grey-box penetration testing

14.7 Let's get brutal

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/cyb_sec_g06

An automotive company has hired someone to conduct both manual and automated penetration testing to make sure that the system is secure from hackers and other malicious attacks. The tester's main role is to discover any weaknesses or vulnerabilities in the company's system that could be exploited by unauthorised users attempting to access the system.

The tester decides to use brute force methods and starts by trying to exploit all the well-known software vulnerabilities to get access to the system infrastructure. This type of test adopts a trial-and-error approach as the tester has no information about the internal workings of the application used by the company, nor about its source code or software architecture. As a result, this particular type of test would take a very long time to complete, so the tester uses automated processes to help uncover the weaknesses and vulnerabilities.

► Based on the above scenario and the approaches used by the tester, which type of penetration test was used?

- ☐ White-box penetration testing
- ☐ Black-box penetration testing
- ☐ Grey-box penetration testing

15 Representation of numbers

Numbers in everyday life are usually represented using the digits 0 to 9, but this is not the only way in which a number can be represented. There are multiple number base systems, which determine which digits are used to represent a number. The number system that we are most familiar with is called **denary** or **decimal (base-10)**, but **binary (base-2)** and **hexa-decimal (hex or base-16)** are also used by computers. You can perform arithmetic calculations on numbers written in other base notations, and even convert numbers between bases.

At a more advanced level, you will learn that representing negative numbers and fractional numbers using binary is also possible. There are several different methods, each with their own advantages and disadvantages.

15.1 Base-2 place values

AQA EDEXCEL EDUQAS OCR WJEC

 isaaccs.org/bq/data_numbases_binary_and_decimal_g01

► The table below has base-2 place values. What is the value of the missing place value?

	16	8	4	2	1
--	----	---	---	---	---

15.2 Converting between binary and denary

AQA EDEXCEL EDUQAS OCR WJEC

 isaaccs.org/bq/data_numbases_binary_and_decimal_g02

► Convert the binary value 01011111 into a denary value.

128	64	32	16	8	4	2	1
0	1	0	1	1	1	1	1

Answer:

15.3 Converting between denary and binary

AQA EDEXCEL EDUQAS OCR WJEC isaacsp.org/bq/data_numbases_binary_and_decimal_g03

► Convert the following denary value into binary.

143

128	64	32	16	8	4	2	1

15.4 Denary to binary - show your workings

AQA EDEXCEL EDUQAS OCR WJEC isaacsp.org/bq/data_numbases_binary_and_denary_g04

► Convert the following denary value into binary.

53

128	64	32	16	8	4	2	1

15.5 Adding two 8-bit binary numbers

AQA EDEXCEL EDUQAS OCR WJEC isaacsp.org/bq/data_numbases_binary_arithmetic_g01

► Add the following two 8-bit binary numbers together and provide your answer in binary.

	0	1	1	0	0	1	1	0
+	0	1	1	1	0	1	0	1
=								

15.6 Binary shifts

AQA

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OCR

WJEC

isaacss.org/bq/data_numbases_binary_arithmetic_g02

► **Multiply** the following 8-bit binary number by **8** and provide your answer in binary.

00001101₂

15.7 The benefits of hexadecimal

AQA

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OCR

WJEC

isaacss.org/bq/data_numbases_hexadecimals_g01

► A friend of yours is learning web development and is interested to know why the colours are written as hexadecimal. They've done some research and think they have worked out why hexadecimal are used.

Help your friend identify which part of their research is **not** accurate.

- ☐ Compared to binary, it is less likely that a digit will be written down incorrectly
- ☐ It's quicker for the processor to interpret as it uses fewer digits
- ☐ It uses fewer digits to represent the same value
- ☐ It is easier to read and interpret

15.8 Binary to hexadecimal conversion 1

AQA EDEXCEL EDUQAS OCR WJEC isaacacs.org/bq/data_numbases_hexadecimals_gcse_g01

► Convert the binary number 01100001 to hexadecimal.

8	4	2	1		8	4	2	1
0	1	1	0		0	0	0	1

- ☐ 61
- ☐ F1
- ☐ A1
- ☐ 6A

15.9 Converting hexadecimal to binary

AQA EDEXCEL EDUQAS OCR WJEC isaacacs.org/bq/data_numbases_hexadecimals_gcse_g02

► Convert the hexadecimal 9B to binary.

8	4	2	1	8	4	2	1

15.10 Hexadecimal to binary 2

AQA EDEXCEL EDUQAS OCR WJEC isaacacs.org/bq/data_numbases_hexadecimals_gcse_g04

► What is the binary equivalent of hexadecimal number F7?

- ☐ 11110110
- ☐ 11111110
- ☐ 11110111

15.11 Binary to hexadecimal conversion 2

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_numbases_hexadecimals_gcse_g03

Convert the binary number 10001111 to hexadecimal.

► Step 1. Split the data into two nibbles and add it to the table below.

8	4	2	1		8	4	2	1

► Step 2. Using the table above, state the hexadecimal value of the **first** nibble.

► Step 3. Using the table above, state the hexadecimal value of the **second** nibble.

15.12 Binary to hexadecimal 2

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_numbases_hexadecimals_gcse_g05

► What is the binary equivalent of hexadecimal number F7?

- ☐ 11111110
- ☐ 01111111
- ☐ 11111111
- ☐ 11110111

15.13 A bit hexed?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_numbases_hexadecimals_gcse_g06

► Complete the table below, hexadecimal is often used by people instead of binary for the following reasons:

Reason	True OR False
It is easier to read and interpret	
It uses more digits to represent the same value	
Compared to binary, it is more likely that the digit will be written down incorrectly	

15.14 8-bit sign and magnitude representation

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_numbases_signed_unsigned_g01

► Using **8-bit sign and magnitude**, convert -40_{10} into binary.

15.15 8-bit two's complement

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_numbases_signed_unsigned_g02

► Using **8-bit two's complement representation**, convert the following binary number into denary.

10001010_2

15.16 What colour is this?

AQA

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EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_data_03

Emily has picked her favourite colour for the background of her website. The value of the colour in binary is $101010101111000011010001_2$ and it is too long to remember! Help Emily by converting the number into hexadecimal.

► Convert the binary number into hexadecimal.

15.17 Zombies

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_data_04

Maya is playing a video game involving zombie enemies. She wants to let her team players know about a trap. Convert the following denary numbers into hexadecimal and combine the answers to find out the secret message.

Secret message: 141_{10} 234_{10} 13_{10}

► Convert the denary numbers into hexadecimal and enter the message as one word, without any blank spaces.

15.18 Putting base-two and base-two together

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/gcse_data_07

What is the result of the binary addition below?

	0	1	1	0	1	1	1	1
+	0	0	1	1	1	0	0	1

► Select the result of the binary addition.

- ☐ 100001100₂
- ☐ 00101001₂
- ☐ 11010110₂
- ☐ 10101000₂

15.19 That's average

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/gcse_data_13

To calculate the arithmetic mean (average) value of a given set of numbers you calculate the sum of values and divide it by the number of values. For example: $(10 + 12 + 14 + 16) / 4 = 13$

The expression used above is written in denary, or base 10. The following expression has been written using numbers from different number bases, the subscript denotes the number base used. Convert each number to denary and then calculate the mean:

► $(10010110_2 + F5_{16} + 7D_{16} + 10110100_2) / 4 = ?$

Give the answer in denary.

15.20 Count the ones

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_data_09

Which of the following base-10 and base-16 numbers has the most bits set to 1 after the number is converted into binary?

► Convert the numbers into binary and select the number that has the most 1s.

- ☐ 100₁₀
- ☐ C5₁₆
- ☐ 79₁₀
- ☐ A2₁₆

15.21 Bonus time

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_data_16

Agneta has just received her latest monthly salary. She has done well this month obtaining two bonus related payments and is putting the money towards her holiday fund.

► The binary numbers below represent Agneta’s bonus payments. Can you work out the total, in binary, of Agneta’s bonus?

Bonus 1				1	0	0	1	0	1	1
Bonus 2		+	1	0	1	0	1	0	0	1
Answer		=								

Pick the correct answer from the options provided.

- ☐ 11110100₂
- ☐ 11100100₂
- ☐ 11100010₂
- ☐ 11110010₂

► What is the total of Agneta’s bonus, in denary?

15.22 Convert it 1

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_data_20

► An unsigned binary integer can be used to represent a positive whole number. Convert the 8-bit unsigned binary integer 01010101 to a denary number.

15.23 Convert it 2

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_data_21

► An unsigned binary integer can be used to represent a positive whole number. Convert the 8-bit unsigned binary integer 10010011 to a denary number.

15.24 Convert it 3

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_data_22

► An unsigned binary integer can be used to represent a positive whole number. Convert the 8-bit unsigned binary integer 00110000 to a denary number.

15.25 Convert it 4

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_data_23

► An unsigned binary integer can be used to represent a positive whole number. Convert the denary number 77 to binary.

15.26 Convert it 5

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_data_24

► An unsigned binary integer can be used to represent a positive whole number. Convert the denary number 100 to binary.

15.27 Convert it 6

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_data_25

► An unsigned binary integer can be used to represent a positive whole number. Convert the denary number 129 to binary.

15.28 Warm colours

Colours on a web page are usually represented in binary by 24-bit codes, i.e. combining 8 bits for a shade of red (R), 8 bits for a shade of green (G), and 8 bits for a shade of blue (B). These long sequences of binary digits can be hard to read, so lookup tables are created which use human-readable descriptions of the colours to index their RGB values, making them more user-friendly.

By way of example, the lookup table below shows four warm colour codes in RGB, 24-bit code binary and hexadecimal. Warm colours generally contain a significant amount of red, mimicking sunlight.

Red is defined by its RGB value of (255,0,0) and its 24-bit binary code value is therefore represented in binary as 111111110000000000000000₂

In hexadecimal, the same colour can be represented by only 6 hexadecimal digits so the colour red is depicted as FF0000₁₆

► Three of the hexadecimal representations are missing from the warm colours table below. Circle the correct list (Red has already been given, as above) of colours in hexadecimal, from the options below, to complete the Hexadecimal table column.

Colour Name	RGB	Binary	Hexadecimal
Red	(255,0,0)	111111110000000000000000 ₂	FF0000 ₁₆
Deep Pink	(255,20,147)	111111110001010010010011 ₂	?
Moccasin	(255,228,181)	111111111110010010110101 ₂	?
Coral	(255,127,80)	111111110111111101010000 ₂	?

Option A

FF0000₁₆

FF1493₁₆

FF7F50₁₆

FFE4B5₁₆

Option B

FF0000₁₆

FF1493₁₆

FF7F50₁₆

FFE4B5₁₆

Option C

FF0000₁₆

FF7F50₁₆

FFE4B5₁₆

FF1493₁₆

Option D

FF0000₁₆

FFE4B5₁₆

FF1493₁₆

FF7F50₁₆

15.29 Adding two binary numbers

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_data_29

When you add two binary numbers, you can use the same method as denary addition: working from the right (the least significant bit), add up the bits in each column with the same place value and, if there is a carry, add it into the next column to the left (that with a higher place value).

► Perform the following calculation and select the option that correctly illustrates the method of adding the two binary numbers.

$11010101_2 + 00010101_2$

☐ Option A

Carry			1	1	1	1	1	1
	1	1	0	1	0	1	0	1

☐ Option B

Carry			1		1		1	
	1	1	0	1	0	1	0	1

☐ Option C

Carry	1	1						
	1	1	0	1	0	1	0	1

☐ Option D

Carry				1		1		1
	1	1	0	1	0	1	0	1

16 Representation of text

All data on a computer system is represented using binary patterns, which are sequences of 1s and 0s. In order to represent text, each individual letter or character must be represented by a unique binary pattern. In order to consistently represent characters from many different languages, these patterns must be agreed throughout the world, through standards such as **ASCII** and **Unicode**. Once you know how a letter is represented, you can calculate the size of a text file and work out how much storage space is required.

16.1 Binary character codes

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/data_rep_text_g01

► The **ASCII character code** for p in binary is 01110000. Using this information, work out the **8-bit character code** in binary for the letter q.

16.2 Denary character codes

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/data_rep_text_g02

► The **ASCII character code** in denary (decimal) for 1 is 49, what would the character code in denary be for the character 3?

16.3 Decoding ASCII

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/data_rep_text_g03

► A word has been encoded with **ASCII character codes** in denary. Decode each character to form a **three-letter word**.

Lower-case letters **a–z** are represented by the ASCII character codes **97 to 122**.

Decode this word: 98 105 116

16.4 Bit-patterns

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/data_rep_text_g04

► How many **bit-patterns** can be created with **4 bits**?

16.5 ASCII count up

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/gcse_data_01

► The standard ASCII code for the character 3 is 51_{10} . What is the ASCII code for 9 in binary?

☐ 00111001_2 ☐ 0101101_2 ☐ 0111001_2 ☐ 00111010_2 ☐ 00001001_2

16.6 ASCII

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_rep_text_g05

► Which **one** of the following statements is **not** correct when discussing the current ASCII character set? Choose the correct answer from the list of three possible answers.

- ☐ **8 bits** are used to store each character in the ASCII character set.
- ☐ The ASCII character set is used to represent text. Each individual letter or character must be represented by a unique binary pattern. These patterns must be agreed throughout the world.
- ☐ **7 bits** are used to represent characters, which allows for a maximum of 128 (2^7) characters to be represented.

16.7 Bits and bit patterns

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_rep_text_g07

The number of bits used in a representation will affect the number of bit-patterns available.

► Add the number of bit patterns into the table to correctly to show how many bits are required to represent them.

	2	3	4	5	6
Num-ber of bit pat-terns					

- ☐ 16
- ☐ 32
- ☐ 8
- ☐ 4
- ☐ 64
- ☐ 2

16.8 Text representation

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/data_rep_text_g09

► Fill in the gaps using the options below.

To represent text digitally, each character needs to have its own _____.
_____ are combinations of 1s and 0s used to represent data inside of a computer. The bit-pattern used for each character becomes a numeric _____.

A collection of characters and the bit-patterns used to represent them is called a _____.

If you are using 4 bits, you can work out the number of bit-patterns by using this formula:

4 bits: $2^4 =$ _____ bit-patterns

You can also take a look at the binary place value headings.

The next place value heading after 3 bits is _____.

- ☐ 8
- ☐ Bit-patterns
- ☐ character code
- ☐ character set
- ☐ unique bit-pattern
- ☐ 16

16.9 ASCII and Unicode differences

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_rep_text_g10

► Use the numbers below to fill in the gaps.

ASCII stands for ‘American Standard Code for Information Interchange’.

It started by using _____ bits to represent characters, which allowed for a maximum of 128 characters to be represented.

These days, _____ bits are used to store each character in the ASCII character set. The first _____ codes in Unicode and ASCII are used to represent the same characters.

Unicode is generally favoured over ASCII because:

- It can support languages with alphabet character sets larger than _____
- Uses up to 32 bits to represent each character and can be used to represent special symbols (e.g. mathematical operators, geometric shapes, arrows, etc).
- Supports over _____ billion possible codes

☐ 128☐ 4☐ 26☐ 7☐ 8

16.10 ASCII and Unicode

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/data_rep_text_g08

► Select the **two** statements that are correct.

- ☐ ASCII is useful for representing characters from languages other than English.
- ☐ Standard ASCII uses 7 bits to represent each character.
- ☐ There is a Unicode character set that uses only 8 bits per character.
- ☐ Unicode can be used to represent up to 256 different characters.
- ☐ Unicode does not include character codes for ideograms.

16.11 Characters

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/data_rep_text_g06

► The bit pattern used for each character becomes a numeric character code. Which of the following are characters?

- ☐ Non-printing commands. For example: Enter, Delete, F1
- ☐ Letters
- ☐ Punctuation symbols. For example: ?/|\£\\$_

17 Representation of images

All data on a computer system is represented using binary patterns, which are sequences of 1s and 0s. In order to represent an image, one method is to store it as if it were a grid of coloured squares, with each colour represented by a unique binary pattern. The image dimensions and the number of colours used are factors that affect the size of the image file.

At a more advanced level, you will learn that images can also be stored as mathematical equations describing shapes, which are then rendered back into an image when viewed by the user. It is useful to know the benefits and drawbacks of each image representation method in order to decide the correct format in which to save a particular image.

17.1 Calculating colours

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_rep_bitmap_g01

A **bitmapped graphic** uses a colour depth of **2 bits** and this allows **4** different colours to be used.

► How many **bits** would be required to represent the colours if a bitmap image included **16** colours?

17.2 How many bytes?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_data_02

► What is the minimum file size in bytes for a 16 x 32 bitmapped graphic which uses 8 colours?

17.3 Game sprites

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/data_rep_bitmap_g02

► Mahira creates a **bitmap graphic** of a character for a game she is making. The image is **8 pixels wide and 8 pixels tall**. How many **pixels** does the image have in total?



► Mahira's bitmap graphic is made up of **four** different colours. To represent **four** different colours, what is the colour depth needed? Give your answer in **bits**.



► What is the **total file size** (excluding any metadata)? Give your answer in **bits**.



17.4 Image files

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/data_rep_bitmap_g04

Fred is searching through the digital photographs he has taken whilst on holiday. When creating a backup of the image files onto an external hard drive, he realises that the file size of each photograph is larger than he expected. This is because, in addition to the pixel data, the file must store additional information so that the image can be reproduced accurately.

► What is the additional information that contains data about data called?

17.5 How many kilobytes?

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_sys_09

Calculate the storage capacity required for a bitmap image with the dimensions 1024 x 768 pixels that has 256 different colours. Then, work out the file size in kB if the file metadata takes up 20% extra space. Present your answer as a real number, including the values after the decimal point.

► What is the storage capacity required? Give your answer in kB.

17.6 Extra colours

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/gcse_data_10

► A bitmapped graphic uses a colour depth of 3-bits and this allows for 8 different colours. If the number of different colours that need to be represented increases to 30, what is the minimum new colour depth required?

17.7 A colourful problem

AQA

EDExcel

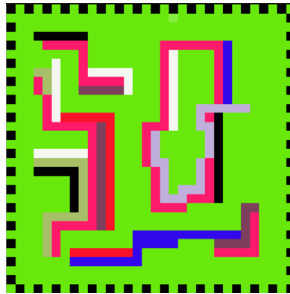
EDUQAS

OCF

WJEC

isaacacs.org/bq/gcse_data_27

Naga has made an icon for a maze solving game she has created. The icon has been created as a bitmapped graphic. It is 32 pixels wide and 32 pixels high and uses 9 different colours.



- How many pixels does the image have in total?

--

- To represent 9 different colours, what is the minimum colour depth needed? Give your answer in **bits**.

--

- What is the total file size (excluding any metadata)? Give your answer in **bits**.

[illegible]

- What is the total file size (excluding any metadata) in **bytes**?

--

17.8 Too much data?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/gcse_data_28

Image files contain metadata which can be described as “data about data”. When Zoe looks at the properties of one of her photos, she sees the following information that has been extracted from the file metadata:

Property	Value
Manufacturer	Canon
Model	EOS 250D
Date and time	2021:04:06 14:31:00
Geolocation	53.61766, -2.1552
Pixel X dimension	2240
Pixel Y dimension	1680
Flash	Flash did not fire

► Which of the items of metadata (from the list shown) is needed to accurately display the image to someone who wants to view the photo? Select all that apply.

- ☐ Date and time
- ☐ Model
- ☐ Flash
- ☐ Pixel X dimension
- ☐ Pixel Y dimension
- ☐ Manufacturer
- ☐ Geolocation

17.9 Image resolution

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_image_bitmap_g03

► Image resolution can be expressed as which **two** of the following?

- ☐ Pixels per inch
- ☐ Bits per pixel
- ☐ Number of pixels across the width and number of pixels in the height
- ☐ The size of each pixel

18 Representation of sound

Sound is a vibration that travels through the air. For a computer to capture sound waves, known as **analogue** sound, it needs to store the sound in a digital format (sequences of 1s and 0s). It also needs to recreate and play back the sound that has been captured. The most common technique used to represent a sound is to take samples (sampling) of the sound at regular intervals. The intervals in which sampling takes place (sample rate), the duration, and the sample resolution (bit depth) all impact the sound quality and size of a file processed by the computer.

18.1 The client's animation

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_sound_01_g

A graphic designer wants to use sound in her client's animation. She has composed a sound that is one minute in length and recorded in stereo sound. The sound was sampled using a 16kHz sampling rate and an 8-bit sample resolution.

► In **bytes**, what storage does this sound require?

☐ 1,920,000

☐ 7,680,000

☐ 15,360,000

☐ 960,000

18.2 Looking for whales and dolphins

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_sound_02_g

A researcher uses a sonar to look for whales and dolphins. The sonar emits a sound signal that makes it possible to detect objects at a reasonable distance underwater. The sonar system operates at a **sampling rate** of 400kHz, with a **sample resolution** of 16 bits.

► What is the **minimum** storage required to store one hour of data received by the sonar in its uncompressed form?

☐ 3TB

☐ 300MB

☐ 300GB

☐ 3GB

18.3 A live performance

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_sound_03_g

► A sound engineer is recording a live performance and uses a sampling rate of 8,000Hz and a sample resolution of 8 bits. What is the minimum file size that will be created for a 30-second recording?

18.4 The event's playlist

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_sound_04_g

► A DJ has created a playlist for an event. She used a sampling rate of **2,000Hz** and a sample resolution of **4 bits** to record each audio file. What is the minimum file size she will need to store a 15-second file recording?

18.5 Annika's ringtone

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_sound_05_g

Annika wants to store a 10-second audio file on her smartphone to use as a ringtone. The audio file was sampled using a 4kHz sampling rate and a 6-bit sample resolution.

► What is the minimum storage required to store Annika's ringtone?

18.6 Jamal's CD

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_sound_06_g

Jamal recorded a CD with a 44,100Hz sampling rate and a sample resolution of **8 bits**.

► How many **bit patterns** does he need to represent each sample?

18.7 Measuring a sound wave

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_sound_07_g

► Shannon converted a sound wave by sampling it. Which of the following terms is being measured when a sound wave passes through a point at certain time intervals?

- ☐ Bit rate ☐ Hertz
☐ Frequency ☐ Amplitude

18.8 File size

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_sound_08_g

► During an exam, you are required to calculate the size of a 10-second mono sound file with a sampling rate of 16,000 hertz and a sample resolution of 8 bits. What would be the size of the file?

18.9 Voice message

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/data_sound_9_g

► Chantelle recorded a 30-second voice message on her mobile phone to send to a friend using a 16-bit sampling resolution and an 8,000Hz sample rate. How many bytes does the recorded message have?

18.10 What does Liu need to know?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_sound_10_g

► Liu is recording a collection of mono audio files to use in his website. He needs to know how much storage is needed to store the files. What information does Liu need to calculate the size of each file? Select all the answers that apply.

- ☐ Sample intervals of the sound
- ☐ Length of the sound
- ☐ Sample resolution of the sound
- ☐ Sampling rate of the sound

18.11 Giga for a gig

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_data_06_v2

► Bob wants to share a sound recording of his favourite gig with his friends. If the performance lasted 1 hour and the recording was made with a sampling rate (frequency) of 15,000 Hz and a sample resolution (size) of 16 bits, what is the minimum size of the audio file in gigabytes (GB)?

18.12 Double trouble

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_data_19

If a music track is recorded in stereo, there are two sets of data (one for each of the left and right channels). A track is recorded in stereo that lasts 3 minutes. The sample resolution is 14 bits and a sampling rate of 55kHz has been used.

► Calculate the minimum stereo storage requirements needed for this stereo music file, in its uncompressed form. Type your answer in **megabytes (MB)**. Assume 1MB is 1,000,000 bytes.

18.13 How many megabytes?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_sys_10

What is the storage capacity required for a song that is 5 minutes long and was recorded using 20,000 samples per second and 8 bits per sample? The file metadata takes up 10% extra space. Give your answer in MBs. Assume 1MB = 1,000,000 bytes.

► What is the storage capacity required? Give your answer in MBs.

19 Compression

Compression is a technique that reduces file size. Some compression techniques are referred to as **lossy**, because they cause data to be lost during the compression process. Other techniques are referred to as **lossless**, because they allow the original data to be exactly reconstructed from the compressed data. Different types of data lend themselves to different compression methods. Popular compression methods include **run length encoding**, **dictionary-based methods**, and **variable length (Huffman) codes**.

19.1 The maze game

AQA

EDEXCEL

EDUQAS

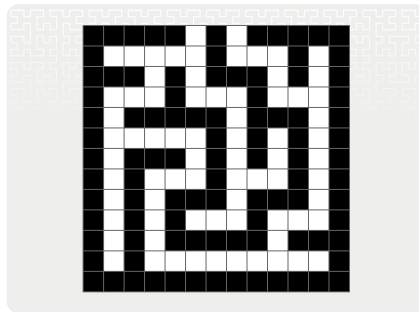
OCR

WJEC

isaaccs.org/bq/data_compr_g02

A game developer is creating a maze game and has used the bitmap below as a background. Let's say that they used RLE to compress the image and used **B** for black and **W** for white. Row 1 would be compressed as B5, W1, B1, W1, B5.

► How would row 12 be compressed?



- ☐ B1, W1, B1, W9, B1
- ☐ B1, W9, B1, W1, B1
- ☐ B1, W1, B1, W1, B5, W1, B3

19.2 Being smart

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/data_compr_g01

A designer is developing a CCTV system that will monitor the flow of traffic and pedestrians at traffic lights throughout a city. The file size of the CCTV video footage is large and needs to be compressed to save on storage space.

► Which compression technique would be most suitable? Take into consideration that the designer needs to significantly reduce storage space and is willing to accept reduced video quality.

☐ Lossless compression

☐ Lossy compression

19.3 Ultrasound machine

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/data_compr_g04

Ultrasound machines are used in the medical field for diagnosis. During a medical examination, part of the sound captured by the ultrasound machine was digitised and converted into the following binary pattern:

0000 0011 1111 1000 0000 0000 0111 1111

► This data is compressed using RLE. Which of the following options would be the correct result of the compression?

☐ 0 0100 0 0010 1 0010 1 0100 1 0001 0 0011 0 0100 0 0100 0 0001 1 0011 1 0100

☐ 0 0110 1 0111 0 1100 1 0111

☐ 0 010010 1 01110

19.4 The correct tree

AQA

isaacacs.org/bq/data_compr_g03

During a test, a student was required to solve a question using Huffman encoding based on this string, which is 20 characters in length:

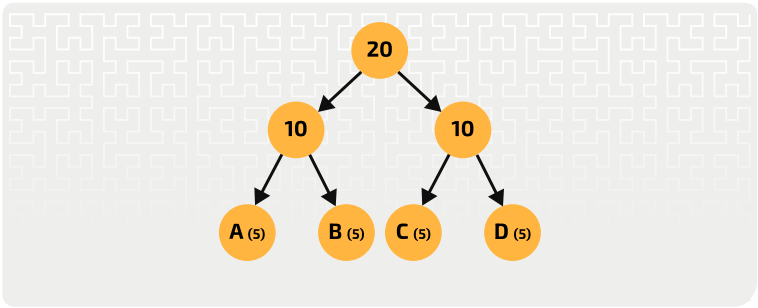
A A B C D A D D C B B A C D B A D C C B

The student was also given this table, which shows the characters, the frequency they appear in the string, and the binary pattern used to represent each character during the compression.

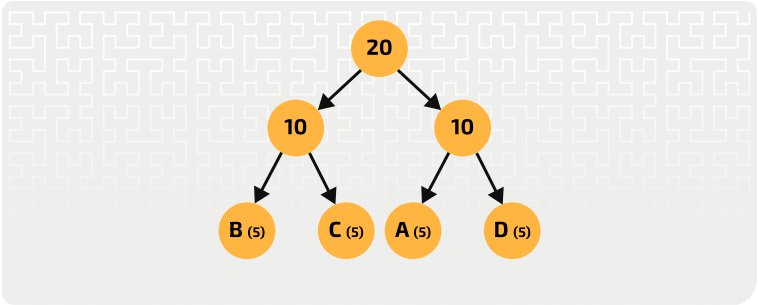
Character	Frequency	Code
A	5	00
B	5	10
C	5	11
D	5	01

► Based on the information provided, which of the following options should the student select to represent the correct Huffman tree for this string?

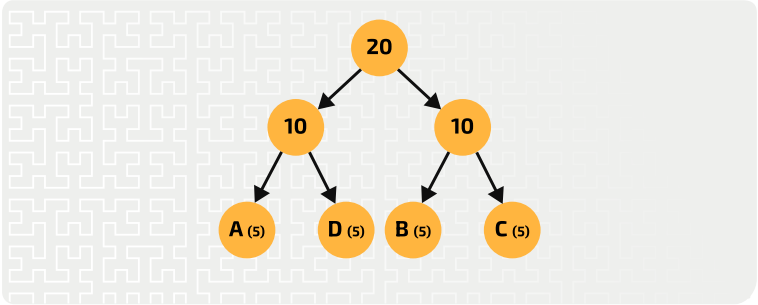
Option 1



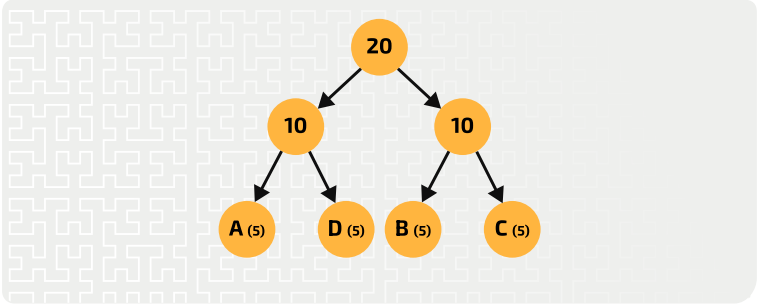
Option 2



Option 3



Option 4



Answer:

19.5 RLE scramble

AQA

EDEXCEL

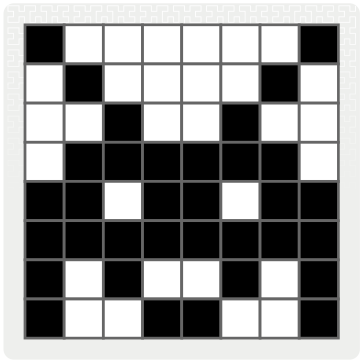
EDUQAS

OCR

WJEC

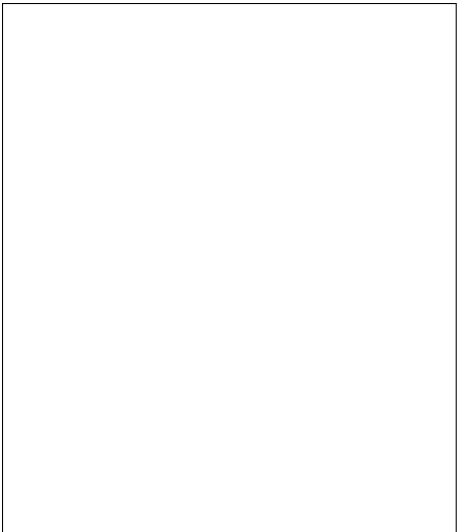
isaacacs.org/bq/data_compr_g05

Black and white pixels are used in the below image to show a bitmap image of a Space Invader.



Run length encoding (RLE) can be used to compress the image. ► Put the encoded lines into the correct order so that they represent the RLE for the image. White is represented with a W and black is represented with a B.

- ☐ B1, W2, B2, W2, B1
- ☐ W2, B1, W2, B1, W2
- ☐ W1, B6, W1
- ☐ B2, W1, B2, W1, B2
- ☐ B8
- ☐ B1, W6, B1
- ☐ B1, W1, B1, W2, B1, W1, B1
- ☐ W1, B1, W4, B1, W1



20 Encryption

Encryption techniques are used to keep data and communications secret from interceptors. **Ciphers** such as the **Caesar cipher** have been used for centuries to **encrypt** and **decrypt** messages using a shared secret known as a **key**. Other cipher techniques are **asymmetric**, which means that the sender and recipient each use different keys to encrypt and decrypt their messages. At a more advanced level, you can explore the **Vernam cipher**, a theoretically perfect cipher in which each character is encrypted using its own key.

20.1 Caesar cipher

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaacacs.org/bq/data_encrypt_g01

You've asked Caesar what he wants to drink, but he doesn't want everyone else to know what he wants. He has sent you the following reply as an encrypted message, along with a key value of 2.

ycvgt

► What does Caesar want to drink?

☐ uyrpc

☐ water

☐ qunyl

☐ juice

20.2 Wrapped caesar cipher

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaacacs.org/bq/data_encrypt_g04

Caesar has received an encrypted message from one of his soldiers. The soldier has sent this message along with a key value of 8.

em vmml eizu cvqnwzua

► What does the soldier want?

☐ mu duut mqhc kdyvehci ☐ we will deal with the enemies

☐ we need warm uniforms ☐ we have been cautious

20.3 Encryption

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_encrypt_g02

► Use the correct key words below to complete the following sentence:

_____ are used to encrypt the data from _____
into _____ and _____ the data from _____
to _____.

☐ plain text☐ decrypt☐ Algorithms☐ cipher text

20.4 Symmetric encryption

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_encrypt_g03

► Use the correct key words below to complete the following sentence:

In a symmetric _____ scheme, the _____ is used
to encrypt and _____ the data.

☐ encryption☐ decrypt☐ same key

20.5 Symmetric encryption

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/data_encrypt_g07

► Read all the sentences below and select one sentence that best describes the keys used for symmetric encryption.

☐ All the keys that are used in symmetric encryption are private.

☐ Three public keys are used when using symmetric encryption.

☐ A symmetric encryption algorithm uses two public keys.

20.6 Encrypt a Caesar cipher

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/data_encrypt_g05

A soldier wants to send this message to Caesar.

more food

The soldier has encrypted and sent this message along with a key value of 7.



► What is the encrypted message?

- ☐ esaq erer
- ☐ pokx fhhw
- ☐ kssy lssd
- ☐ tvyl mvvk

20.7 Asymmetric encryption

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/data_encrypt_g06

► Read all the sentences below and select one sentence that best describes the keys used for asymmetric encryption.

- ☐ All the keys that are used in asymmetric encryption are private.
- ☐ An asymmetric encryption algorithm uses a pair of keys: a public key and a private key.
- ☐ Three public keys are used when using asymmetric encryption.

21 Database concepts

Data is fundamental to any computer system. With data comes the need for a system that is able to store, sort, and search through it. Imagine the amount of users a popular social media company has, then think about storing every post they have made and the media that comes with it, such as a video, image, or animated GIF. Without a database, you wouldn't be able to log in to access your posts, let alone be able to make connections with other users and see their posts. In this section you will learn that databases can be as simple as a single file, and as complex as a set of related **tables**, each with **fields** and **keys**. It is important that data within a database is kept up-to-date and consistent, and it is crucial to avoid storing the same data in multiple places. At a more advanced level, you will learn that creating an **entity relationship model** and designing a set of tables using the rules of **normalisation** is important. Challenges arise when lots of users want to access data simultaneously, and you will learn that **transactions** and **transaction processing** are needed to ensure that the data does not get out of sync.

21.1 Foreign keys

AQA

OCR

isaaccs.org/bq/data_dbs_concepts_g02

► Which of the following options provides the correct definition of a foreign key?

- ☐ A field that uniquely identifies a record in a table
- ☐ A field that is used to link to another table through a matching value in a primary key field
- ☐ A term used to describe any field in a database table that does not contain unique data.

21.2 Sports database

AQA

OCR

isaacss.org/bq/data_dbs_concepts_g01

A local sports league needs to keep a record of the name of players and which team they play for. Their database currently has three tables:

- The table **Player** contains the following fields: **PlayerID**, **FirstName**, **Surname**, **DateOfBirth**, **Address**, **TelephoneNumber**
- The table **Team** contains the following fields: **TeamID**, **TeamName**, **KitColour**
- The table **Registrations** contains the following fields: **RegistrationID**, **PlayerID**, **TeamID**, **RegistrationDate**

► Which field in the **Player** table is a **primary key**?

- ☐ Address
- ☐ PlayerID
- ☐ DateOfBirth
- ☐ FirstName

► Which fields in the **Registrations** table are **foreign keys**?

- ☐ PlayerID, TeamID
- ☐ RegistrationID, RegistrationDate
- ☐ RegistrationID, PlayerID, TeamID

21.3 Flat file databases

AQA

isaacccs.org/bq/data_dbs_concepts_g04

Amelie provides a dog walking service and keeps all of the details of her customers in a flat file database. A sample of the data she holds is below:

Customer-Name	Address	TelephoneNumber	Dog-Name
Frank Zanetti	18 Gravel Lane, Banks	0778495743	Bobby
Bella Talbot	89 Carr Lane, Chorley	0782347512	Ant
Bella Talbot	89 Carr Lane, Chorley	0782347512	Dec
Jenna Alexander	10 St Johns Green, Leyland	0797748321	Bruno
Jenna Alexander	10 St Johns Green, Leyland	0797748321	Alfie
Jenna Alexander	10 St Johns Green, Leyland	0797748321	Rover

► From the choices below, choose the term that best describes the problem with the data in this flat file database.

- ☐ Data inconsistency
- ☐ Data integrity
- ☐ Data redundancy

22 SQL

SQL is short for **Structured Query Language**; it's a language that is used for constructing statements called **queries** to perform operations on the data within a database. SQL operations include:

- Searching and retrieving records
- Inserting new records into the database tables
- Deleting records and individual items of data from tables
- Updating existing data by replacing it with new data

22.1 Insert the download

AQA

isaacacs.org/bq/data_dbs_dml_g01

Below is a table showing app downloads for a customer.

Table AppDownloads

DownloadID	App	PurchaseDate	Price
187362	Monopoly Classic	2022-03-01	4.99
273340	Mini Football	2022-03-20	2.79
317333	Stardew Valley	2021-11-19	4.99
387737	Game Dev Tycoon	2021-04-28	4.99
622100	Alien Isolation	2022-02-12	12.99

► The customer has recently downloaded another app called **Flow Free**. Use the options below to complete the following query that will insert the new download into the table.

```
1 _____ INTO _____ (DownloadID, App,
    PurchaseDate, Price)
2 _____ (931408, _____, '2022-14-18',
    0.00) _____
```

- ☐ INSERT
- ☐ AppDownloads
- ☐ VALUES
- ☐ 'Flow Free'
- ☐ ;

22.2 Property price update

AQA

isaacacs.org/bq/data_dbs_dml_g02

An estate agents in the north east of England has a database of properties that are currently for sale. An extract from the **Properties** table is below. The owner of the property at **8 New Elvet** wants to change the list price of their property.

PropertyID	Address	City	PropertyType	Bed-rooms	List-Price
Dur032	8 New Elvet	Durham	detached	3	350000
Dur070	2 Sherburn Rd	Durham	terraced	2	200000
New635	6A Salters' Rd	Newcastle Upon Tyne	semi-detached	3	220000
New388	29 Leazes Rd	Newcastle Upon Tyne	terraced	4	480000
Sun112	2 Roker Lane	Sunderland	detached	4	450000

► Complete the SQL query below using the words provided to update the **ListPrice** of the property at **8 New Elvet**, **Durham** to **330000**.

```
----- Properties
<br>
----- ListPrice = 330000
<br>
----- PropertyID = 'Dur032';
<br>
```

- ☐ UPDATE
- ☐ INSERT
- ☐ WHERE
- ☐ SET

22.3 DELETE students

AQA

isaacacs.org/bq/data_dbs_dml_g03

A local school PE department stores data in a database about students taking part in sports day. Some of the students are no longer taking part.

An extract from the table **Students** is below.

FirstName	LastName	Team	YearGroup
Charlie	Heard	Red	11
Amisha	Patel	Blue	11
Jenny	Jackson	Green	10
Viran	Ahmed	Green	9
Jaden	Beckford	Blue	10

► When the following SQL query is executed, how many records will be deleted?

```
1 DELETE FROM Students
2 WHERE Team = 'Blue' AND YearGroup = 11;
```

- ☐ 3 records
- ☐ 2 records
- ☐ 1 record

22.4 Construct a select query

AQA OCR

isaacss.org/bq/data_dbs_dql_g01

A local school PE department stores data in a database about students taking part in sports day.

An extract from the table **Students** is below:

FirstName	LastName	Team	YearGroup
Charlie	Heard	Red	11
Amisha	Patel	Blue	11
Jenny	Jackson	Green	10
Viran	Ahmed	Green	9
Jaden	Beckford	Blue	10

A member of staff wants a list of all of the students' **FirstName**, **LastName**, and **Team** that are in the database table.

```
SELECT _____, _____, _____  
  
_____ Students_____
```

► Fill in the gaps above using the below options to complete the query:

- ☐ *
- ☐ ;
- ☐ LastName
- ☐ Team
- ☐ YearGroup
- ☐ FROM
- ☐ WHERE
- ☐ FirstName

22.5 SELECT, FROM, WHERE

AQA

OCR

isaacacs.org/bq/data_dbs_dql_g02

A local school PE department stores data in a database about students taking part in sports day.

An extract from the table **Students** is below:

FirstName	LastName	Team	YearGroup
Charlie	Heard	Red	11
Amisha	Patel	Blue	11
Jenny	Jackson	Green	10
Viran	Ahmed	Green	9
Jaden	Beckford	Blue	10

A member of staff needs to know which Year 11 students are taking part. Tick **one** box in each of the sections below to identify the correct SQL statement to select the names and team for all year 11 students.

► Section 1

- ☐ SELECT FirstName, LastName, Team
- ☐ SELECT FirstName AND LastName AND Team
- ☐ SELECT *
- ☐ SELECT FirstName, LastName, Team, YearGroup

► Section 2

- ☐ FROM database
- ☐ FROM students
- ☐ FROM Students
- ☐ FROM table Students

► Section 3

- ☐ WHERE YearGroup = 11;
- ☐ WHERE YearGroup = "Year 11";
- ☐ WHERE YearGroup == 11;

22.6 Top films

AQA

isaacacs.org/bq/data_dbs_dql_03

Aleema has bookmarked the following films on a streaming service she subscribes to.

FilmID	Title	Certificate	Duration	Genre	Rating
1	Hamilton	12	160	History	8.9
2	Spirited Away	PG	125	Animation	8.6
3	Back to the Future	PG	116	Adventure	7.5
4	Coco	PG	105	Animation	8.4
5	Raiders of the Lost Arc	PG	115	Adventure	7.3
6	Toy Story	U	81	Animation	9.1

SELECT Title, Certificate, Duration, Genre, Rating

FROM Films

ORDER BY -----;

► Complete the **SELECT** query above so that the films are ordered from the highest rating to the lowest and Aleema knows which film to watch first.

- ☐ FilmID
- ☐ DESC
- ☐ Rating
- ☐ ASC

22.7 App downloads

AQA OCRisaacacs.org/bq/data_dbs_dql_g04

Dana has been looking at her app purchases. She wants to redownload an app that she thinks she downloaded in November 2021. An extract from the `Downloads` table is below:

App	PurchaseDate	Price
Monopoly Classic	2022-03-01	4.99
Mini Football	2022-03-20	2.79
Stardew Valley	2021-11-19	4.99
Game Dev Tycoon	2021-04-28	4.99
Alien Isolation	2022-02-12	12.99

► Choose the option that would complete the query below to help Dana find the game.

```
1 SELECT *
2 FROM Downloads
3 WHERE _____ ;
```

- ☐ PurchaseDate > '2021-11-01' OR PurchaseDate < '2021-11-30'
- ☐ PurchaseDate > '2021-11-01' AND < '2021-11-30'
- ☐ PurchaseDate > '2021-11-01' AND PurchaseDate < '2021-11-30'

22.8 Location, location

AQA OCR

isaaccs.org/bq/data_dbs_dql_g05

An estate agent in the north east of England has a database of properties that are currently for sale. An extract from the **Properties** table is below:

PropertyID	Address	City	PropertyType	Bedrooms	ListPrice
Dur032	8 New Elvet	Durham	detached	3	350000
Dur070	2 Sherburn Rd	Durham	terraced	2	200000
New635	6A Salters' Rd	Newcastle Upon Tyne	semi-detached	3	220000
New388	29 Leazes Rd	Newcastle Upon Tyne	terraced	4	480000
Sun112	2 Roker Lane	Sunderland	detached	4	450000

A buyer is using the agents' website to find all available properties in **Sunderland**.

► Use the lines of code below to construct a SQL query that would return the following fields for the buyer:

Address, City, PropertyType, Bedrooms, ListPrice.

Not all lines of code provided are required to solve the problem

- ☐ SELECT *
- ☐ WHERE City = 'Sunderland';
- ☐ FROM Properties
- ☐ WHERE 'City' = 'Sunderland'
- ☐ SELECT Address, City, PropertyType, Bedrooms, ListPrice
- ☐ FROM table

--

► Using the table above, which of the properties in the list below would be returned when the follow query is executed:

```
1  SELECT PropertyID
2  FROM Properties
3  WHERE (Bedrooms = 3 OR Bedrooms = 4) AND ListPrice <= 350000;
```

- ☐ D032, New388, Sun112
- ☐ New635
- ☐ D032, New635

23 Data structures

Almost all programs process data in some form. Data needs to be stored in a way that means it can be easily manipulated by program code. Very basic (**primitive**) data types, such as an integer (a whole number) or Boolean (True or False), allow you to store a single value. It is possible to combine these primitive data types to create **compound data types** (a record) or to create more complex data structures (arrays and lists).

At a more advanced level, **static** and **dynamic data structures** are introduced. Together with **abstract data types**, such as stacks and queues, they allow the more sophisticated organisation and manipulation of data, and provide methods for adding, removing, and traversing data.

23.1 Two-dimensional array

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/dsa_datastruct_g01

This two-dimensional array shows the data structure for storing words for schoolchildren to learn their spellings.

Index	0	1	2	3
0	school	pull	where	here
1	path	floor	sugar	bread
2	accident	answer	eight	critical

After the array has been created and saved as **spelling_words** in a program, the following line of code is run:

```
PRINT(spelling_words[2,1])
```

► When the first index is the row number and the second index is the column number, what will be the output on the screen?

- ☐ answer
- ☐ path
- ☐ sugar

23.2 Dictionary

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g02

A dictionary is a _____ structure that stores data using a _____ and value system. You can store any type of object in a dictionary, and when presented with a _____, the dictionary will return the value associated with that _____.

► Use the words below to complete the above paragraph.

☐ key☐ value☐ data

23.3 Arrays

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g03

An array is a data _____ that holds a number of data _____ or elements of the _____ data type. When you want to store many pieces of data that are related, it is often better to use an array _____ of many variables.

► Use the words below to complete the above paragraph.

☐ structure☐ items☐ same☐ instead

23.4 Index numbers

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g04

This one-dimensional array stores different weather conditions:

Sunny	Windy	Cloudy	Raining	Snowy
-------	-------	--------	---------	-------

► Assuming the array is indexed from 0, what weather condition is in index position 2?

23.5 Static data

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g05

► When discussing static data structures, which **one** of the following statements is **incorrect**?

- ☐ Memory is not wasted. The size of the array is declared for the number of data items that will be added when the program is run.
- ☐ If insufficient memory space has been allocated, the program may crash or not work properly.
- ☐ Memory is not wasted, for example, the elements are only declared for the same amount of data items when the program is run.

23.6 Dynamic data

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g06

► Is the following statement **true** or **false**?

Dynamic data structures are memory efficient. The memory capacity of a dynamic data structure is not fixed, and the number of data items that it can hold is constrained only by the overall memory allocation to the program. A list is an example of a dynamic data structure.

- ☐ True
- ☐ False

23.7 How many students?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/gcse_prog_18

In a college, two groups of students were created, group1 and group2. Mr Patel (the English teacher) asked Mrs Smith (the Computer Science teacher) how many students are there in group 1 and 2 when the groups are added together.

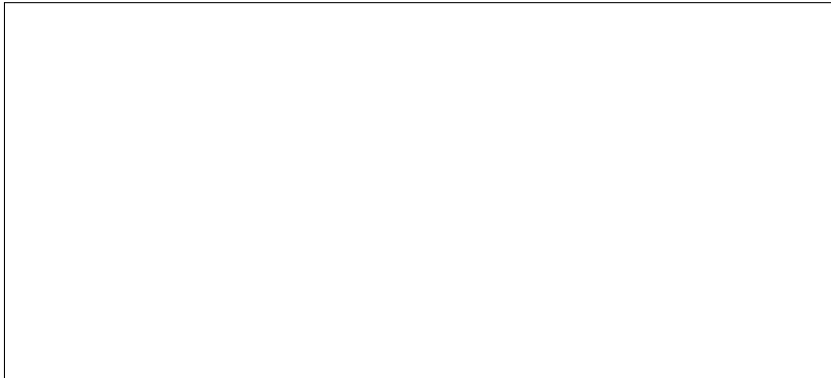
Mrs Smith loves to challenge people, so she gave the algorithm below as her answer and told Mr Patel to use it to find the number of students.

- group1 is an array of students with indexes from 0 to 17
- group2 is an array of students with indexes from 0 to 19

Remember that the `LEN()` function will return the total number of items in an array.

```
1 FUNCTION total_students(group1, group2)
2     total = LEN(group1) + LEN(group2)
3     RETURN total
4 ENDFUNCTION
```

► What answer will be given by running the pseudocode provided by Mrs Smith?



23.8 That's not correct!

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g07

► Data structures can be static or dynamic. Which of the following statements is **incorrect**?

- ☐ A static array cannot be resized at runtime.
- ☐ A dynamic data structure is a collection of data in memory that can grow or shrink in size at runtime.
- ☐ The elements of a dynamic linked list can be accessed directly by index.
- ☐ A static array uses an allocation of contiguous (one after another) locations in main memory.

23.9 One- and two-dimensional arrays

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g09

► Fill in the gaps using the options below.

A one-dimensional array is a data structure that contains a set of elements of the same _____. To access an element in an array, you use an _____. This specifies the position of the element in the array.

To access an element in a _____, you need to specify a column number and a _____ number. These are known as a column index and a _____.

- ☐ two-dimensional array
- ☐ row index
- ☐ index number
- ☐ row
- ☐ data type

23.10 Describe lists and dictionaries

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_datastruct_g08

► Fill in the gaps using the options below.

A list is a _____ collection of data items in some sort of order, which means that each _____ occupies a specific _____ in the list. A list is a ‘built-in’ data structure (or data type) in many programming languages.

In the programming world, a dictionary is a data structure that is used to store data using a key and value system, just like a paper copy of a dictionary. The word you are looking for is the _____ and the definition is the _____. When a dictionary is presented with an obvious _____, the dictionary will return the _____ associated with that _____.

- ☐ key
- ☐ value
- ☐ position
- ☐ element
- ☐ term
- ☐ linear

24 Searching algorithms

Computers need to search through lists of data all the time; for example, when trying to find a file with a particular name on your computer, or when using a search engine to find websites on the internet that match certain keywords.

When faced with the problem of **searching** for an item in a list of items, you may have to deal with an **unordered list**. With an unordered list, the only way to find a specific item is to look at every item in the list one after another. This algorithm is called a **linear search**. If the list of items is **ordered**, you can repeatedly split the list in half and check only the half in which the item that you are searching for might be found. This algorithm is called a **binary search**.

At a more advanced level, you will learn that these searching algorithms can be implemented either **iteratively** or **recursively**. You will learn that a **binary search tree** could also be used. You will also learn that the relative **complexity** of each algorithm can be compared, to allow the best choice to be made for a specific problem.

24.1 Why linear?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/dsa_search_g10

► Identify **three** reasons why you may choose to perform a linear search rather than a binary search on a list of data.

- ☐ The list is unsorted
- ☐ The list is very short
- ☐ The list is very long
- ☐ The list is sorted
- ☐ The algorithm is simpler to implement

24.2 Search for knitwear sales

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/dsa_search_binary_g02

Tom sells knitwear via his website. He has a list with the number of products that he’s sold in the last ten days. He wants to find out if there is a day that he sold 22 products.

products_sold									
11	16	17	22	25	30	35	46	57	72

Tom is using the binary search algorithm to search through the list of sales.

The binary search implementation uses the formula `midpoint = (first + last)DIV 2` to calculate the midpoint position. Remember that integer or floor division (DIV) is when any remainder of a division is discarded. For example, $11 \div 5 = 2$ remainder 1 and so 11 DIV 5 will return 2.

► Fill in the blanks with the values provided below to complete the steps that a binary search will follow when searching for 22 in the `products_sold` list.

Take the ordered list `products_sold` and the search item 22.

Initially, set the range of items where the search item might be found to be the entire list.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is greater than 22.

Change the range to focus on the items before the midpoint.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is less than 22.

Change the range to focus on the items after the midpoint.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is _____ 22.

Change the range to focus on the items _____ the midpoint.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is equal to 22.

The search item was found in the list, stop searching.

- ☐ less than
- ☐ 22
- ☐ greater than
- ☐ 25
- ☐ before
- ☐ 30
- ☐ after
- ☐ 35
- ☐ 11
- ☐ 46
- ☐ 16
- ☐ 57
- ☐ 17
- ☐ 72

Tom sells knitwear via his website. He has a list with the number of products that he’s sold in the last ten days. He wants to find out if there is a day that he sold 35 products.

products_sold									
11	16	17	22	25	30	35	46	57	72

Tom is using the binary search algorithm to search through the list of sales.

The binary search implementation uses the formula `midpoint = (first + last)DIV 2` to calculate the midpoint position. Remember that integer or floor division (DIV) rounds down the result of a calculation to the nearest integer. For example, 11 DIV 5 will return 2.

► Fill in the blanks with the values provided below to complete the steps that a binary search will follow when searching for 35 in the `products_sold` list.

Take the ordered list `products_sold` and the search item 35.

Initially, set the range of items where the search item might be found to be the entire list.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is less than 35.

Change the range to focus on the items after the midpoint.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is _____ 35.

Change the range to focus on the items _____ the midpoint.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is _____ 35.

Change the range to focus on the items _____ the midpoint.

Find the item at the midpoint position: _____.

Compare the item at the midpoint to the search item: the item at the midpoint is equal to 35.

The search item was found in the list, stop searching.

- | | |
|---------------------------------------|-----------------------------|
| ☐ less than | ☐ 22 |
| ☐ greater than | ☐ 25 |
| ☐ before | ☐ 30 |
| ☐ after | ☐ 35 |
| ☐ 11 | ☐ 46 |
| ☐ 16 | ☐ 57 |
| ☐ 17 | ☐ 72 |

24.3 Search for a book

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/dsa_search_binary_g03

Sally has a blog about children’s books. She has a list with the top seven children’s books that were chosen by her readers. You can see the items of the list below:

top7_books						
Animal Farm	Black Beauty	Elmer	Heidi	Robin-son Crusoe	The Little Prince	Winnie-The-Pooh

Sally wants to find out if the book ‘Elmer’ is in the list. She is using the binary search algorithm to search through the list of books.

The binary search implementation uses the formula `midpoint = (first + last)DIV 2` to calculate the midpoint position. Remember that integer or floor division (DIV) is when any remainder of a division is discarded. For example, $11 \div 5 = 2$ remainder 1 and so 11 DIV 5 will return 2.

► Fill in the blanks with a suitable word from the ones provided below to complete the steps that a binary search will follow when searching for **Elmer** in the **top7_books** list.

Take the ordered list **top7_books** and the search item **Elmer**.

Initially, set the range of items where the search item might be found to be the entire list.

Find the item at the midpoint position: _____ is the midpoint item.

Compare the item at the midpoint to the search item: the item at the midpoint is greater than **Elmer**.

Change the range to focus on the items before the midpoint.

Find the item at the midpoint position: _____ is the midpoint item.

Compare the item at the midpoint to the search item: the item at the midpoint is _____ **Elmer**.

Change the range to focus on the items _____ the midpoint.

Find the item at the midpoint position: _____ is the midpoint.

Compare the item at the midpoint to the search item: the item at the midpoint is equal to **Elmer**.

The search item was found in the list, stop searching.

- ☐ less than
- ☐ greater than
- ☐ before
- ☐ after
- ☐ Animal Farm
- ☐ Black Beauty
- ☐ Elmer
- ☐ Heidi
- ☐ Robinson Crusoe
- ☐ The Little Prince
- ☐ Winnie-The-Pooh

Sally has a blog about children’s books. She has a list with the top seven children’s books that were chosen by her readers.

top7_books						
Animal Farm	Black Beauty	Elmer	Heidi	Robinson Crusoe	The Little Prince	Winnie-The-Pooh

► Sally is using the binary search algorithm to find out if the book **Peter Pan** is in the list. How many comparisons will the binary search algorithm perform before it finishes?

24.4 Why not binary?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/dsa_search_g04

You have been provided with some cards that are in the following order:



- Why could you **not** use a binary search for this data?
- ☐ The data is not a list of numbers.
 - ☐ The data is not sorted in an ascending (low to high) or descending (high to low) order.
 - ☐ There is not enough data.

24.5 Binary or linear?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/dsa_search_g05

You have been provided with some cards that are in the following order:



- You need to find out if the number 4 is in the list of cards. Which searching algorithm would require the **fewest** number of comparisons to find the data?
- ☐ Linear
 - ☐ Binary
 - ☐ Both would be the same
- You now need to find out if the number 9 is in the list of cards. Which searching algorithm would require the **fewest** number of comparisons to find the data?
- ☐ Linear
 - ☐ Binary
 - ☐ Both would be the same

24.6 Efficient searching

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_search_g06

► Which of these is the **most** efficient implementation of a linear search?

Option A

```
1 FUNCTION linear_search(items, search_item)
2
3     // Initialise the variable
4     found_index = -1
5
6     // Repeat until the end of the list is reached
7     FOR current = 0 TO LEN(items) - 1
8
9         // Compare the item at the current index to the
           search item
10        IF items[current] == search_item THEN
11            // If the item has been found, store the current
               index
12            found_index = current
13        ENDIF
14    NEXT current
15
16    // Return the index of the search_item or -1 if not found
17    RETURN found_index
18 ENDFUNCTION
```

Option B

```
1 FUNCTION linear_search(items, search_item)
2
3     // Initialise the variables
4     found_index = -1
5     current = 0
6     found = False
7
8     // Repeat while the end of the list has not been reached
9     // and the search item has not been found
10    WHILE current < LEN(items) AND found == False
11
12        // Compare the item at the current index to the
           search item
13        IF items[current] == search_item THEN
14            // If the item has been found, store the current
               index
15            found_index = current
```

```
16         found = True // Raise the flag to stop the loop
17     ENDIF
18     current = current + 1 // Go to the next index in the
        list
19 ENDWHILE
20
21 // Return the index of the search_item or -1 if not found
22 RETURN found_index
23
24 ENDFUNCTION
```

Answer:

24.7 Binary comparisons

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaaccs.org/bq/dsa_search_g07

A local hockey club keeps a directory of players' email addresses, which they use to send out information about training and fixtures. It currently has 1,000 entries.

► What is the maximum number of comparisons a binary search would have perform to find a specific player's email address?

24.8 Binary tracing

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_search_g09

The following list of items is stored in an array:

items							
index	0	1	2	3	4	5	6
value	2	4	5	8	12	15	18

The following algorithm has been coded and will be used to search for the number 12 as highlighted in the array.

```

1 FUNCTION binary_search(items, search_item)
2
3     // Initialise the variables
4     found = False
5     found_index = -1
6     first = 0
7     last = LEN(items) - 1
8
9     // Repeat while there are still items between first and
10    last
11    // and the search item has not been found
12    WHILE first <= last AND found == False
13
14        // Find the midpoint position (in the middle of the
15        range)
16        midpoint = (first + last) DIV 2
17
18        // Compare the item at the midpoint to the search
19        item
20        IF items[midpoint] == search_item THEN
21            // If the item has been found, store the
22            midpoint position
23            found_index = midpoint
24            found = True // Raise the flag to stop the loop
25
26        // Check if the item at the midpoint is less than
27        the search item
28        ELSEIF items[midpoint] < search_item THEN
29            // Focus on the items after the midpoint
30            first = midpoint + 1
31
32        // Otherwise the item at the midpoint is greater
33        than the search item

```

```
28         ELSE
29             // Focus on the items before the midpoint
30             last = midpoint - 1
31         ENDIF
32     ENDWHILE
33
34     // Return the position of the search_item or -1 if not
35     found
36     RETURN found_index
37 ENDFUNCTION
```

► Complete the trace table for the algorithm by adding the options provided into the correct positions.

found_ index	first	last	mid- point	items [midpoint]	found	return value
-1	0	6	3	8	false	
	4					
	4	_____	2	_____	_____	
		4				
4	4	4	_____	12	_____	

- ☐ True
- ☐ False
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 8
- ☐ 15

24.9 Search for a basketball score

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/dsa_search_linear_g01

Amaia coaches a school basketball team. She keeps the scores that the team got in the latest tournament in a list called `basketball_finals`. You can see the items of the list below:

basketball_finals							
120	250	101	150	80	95	147	165

Amaia is using the linear search algorithm to find out if the team scored 150 points in any of the games.

► Put in order the steps that the linear search algorithm will follow when searching for the item 150 in the list `basketball_finals`.

- ☐ Go to the next item in the list: 250
- ☐ Go to the next item in the list: 150
- ☐ Compare 120 to 150: 120 is not equal to 150
- ☐ Compare 250 to 150: 250 is not equal to 150
- ☐ Compare 150 to 150: 150 is equal to 150
- ☐ The item was found, stop searching
- ☐ Go to the next item in the list: 101
- ☐ Start from the first item in the list: 120
- ☐ Compare 101 to 150: 101 is not equal to 150

Amaia coaches a school basketball team. She keeps the scores that the team got in the latest tournament in a list called `basketball_finals`. You can see the items of the list below:

basketball_finals							
120	250	101	150	80	95	147	165

► Amaia is using the linear search algorithm to find out if the team scored 200 points in any of the games. How many comparisons will the linear search algorithm perform before it finishes?

24.10 Midpoint

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/dsa_search_g08

You have been given a set of cards that are in the following order:



► A binary search is going to be used to search for an item in the list. Which card is at the first midpoint calculated by the algorithm?

25 Sorting algorithms

Sorting data is an essential data management task. You may need to display information in a specific order, or you may need to sort your data to make it quicker to search for something within the data set. There are many standard algorithms that can be used to sort data, such as **bubble sort** and **merge sort**.

At a more advanced level, you will learn that these sorting algorithms can be implemented in code in a variety of ways. You will also learn that the relative **complexity** of each algorithm can be compared, to allow the best choice to be made for a specific problem.

25.1 Sorting basketball scores

AQA

EDUQAS

OCR

WJEC

isaaccs.org/bq/dsa_sort_g01

Amaia coaches a school basketball team. She keeps the scores that the team got in the latest tournament in a list called `basketball_finals`. You can see the items of the list below:

basketball_finals				
250	120	95	101	80

► Amaia wants to use the merge sort algorithm to sort the scores from lowest to highest value. Fill in the gaps with the values to show the order of the items after the **first merge**. Assume that the splitting stage has already completed and each value is in a list of its own.

Final split				
250	120	95	101	80
Merge 1				

- ☐ 80
- ☐ 95
- ☐ 101
- ☐ 120
- ☐ 250

25.2 Inserting countries

EDEXCEL OCR

isaacss.org/bq/dsa_sort_g02

Geraint is using the **insertion sort** algorithm to put their list of top five countries to visit into alphabetical order, from A to Z.

countries				
Peru	Laos	Fiji	Egypt	Cuba

The insertion sort algorithm has been started on the list below. In the first row, the first item is highlighted to show it is in the sorted sublist and the unsorted sublist contains the remaining items.

In the second row, the first pass has completed and the sorted sublist now contains two items, which are highlighted.

► Fill in the blanks to show the order of the countries at the end of each pass when performing an **insertion sort**. Use a new row for each pass.

countries				
Peru	Laos	Fiji	Egypt	Cuba
Laos	Peru	Fiji	Egypt	Cuba

- ☐ Cuba
- ☐ Egypt
- ☐ Fiji
- ☐ Laos
- ☐ Peru

25.3 Improving bubble sort

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_sort_g04

► A bubble sort algorithm is written in pseudocode below. Study the pseudocode and then select two changes that could be made to improve the efficiency of the algorithm.

```
1  PROCEDURE bubble_sort(items)
2      num_items = LEN(items)
3      FOR pass_num = 1 TO num_items - 1
4          FOR index = 0 TO num_items - 2
5              IF (items[index] > items[index + 1]) THEN
6                  temp = items[index]
7                  items[index] = items[index + 1]
8                  items[index + 1] = temp
9              ENDIF
10         NEXT index
11     NEXT pass_num
12 ENDPROCEDURE
```

- ☐ The inner for loop could be changed so that the number of repetitions is reduced by 1 after each pass
- ☐ The inner for loop could be changed to a while loop that only swaps items if they are out of order
- ☐ The variable **temp** is not needed when swapping items in this way and could be removed
- ☐ The outer for loop could be changed so that the number of swaps made is reduced by 1 after each pass
- ☐ The outer for loop could be changed to a while loop that stops once no swaps are made during a single pass

25.4 Bubble sort puzzle

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/dsa_sort_g05

► Put the lines of code provided into the correct order to create a bubble sort algorithm. Please note that you should use correct indentation in your answer.

- ☐ NEXT pass_num
- ☐ FOR index = 0 to num_items - 2
- ☐ IF (items[index] > items[index + 1]) THEN
- ☐ num_items = LEN(items)
- ☐ NEXT index
- ☐ END IF
- ☐ items[index] = items[index + 1]
- ☐ items[index + 1] = temp
- ☐ FOR pass_num = 1 TO num_items - 1
- ☐ temp = items[index]

25.5 Where to start an insertion sort?

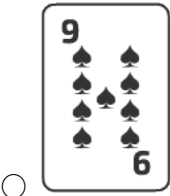
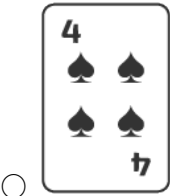
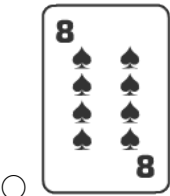
EDEXCEL OCR

isaacacs.org/bq/dsa_sort_g08

You have been given some playing cards that have not been sorted:



► In an insertion sort, which would be the first card in the **unsorted** sublist and therefore the **first** card to insert?



25.6 Bubbling countries

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/dsa_sort_g03

Geraint is using the **bubble sort** algorithm to put their list of top five countries to visit into alphabetical order, from A to Z.

► Using the **countries** list below, fill in the blanks to show the order of the countries after **each pass of the bubble sort algorithm**, using a new row for each pass. The algorithm will pass over the entire list four times.

countries				
Laos	Egypt	Peru	Fiji	Cuba

- ☐ Cuba
- ☐ Egypt
- ☐ Fiji
- ☐ Laos
- ☐ Peru

26 Legislation

Until the **Computer Misuse Act 1990**, there were no legal provisions in UK law governing the use and misuse of computer systems. However, the use of computers has become so widespread that further laws such as the **Regulation of Investigatory Powers Act 2000** have been enacted to take into account technological growth and the widespread use of encryption. There have also been recent changes to data protection law. The **General Data Protection Regulation (GDPR)** in EU law creates a European standard to protect individuals, and regulates how companies and organisations may collect, hold, and use personal data. The **Data Protection Act 2018** is the UK's implementation of the GDPR. Legislation also exists around the use of copyrighted material through the **Copyright, Designs and Patents Act 1988**.

26.1 Which licence type?

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaacss.org/bq/iot_legal_g01

► Select the type of licence that allows software code to be studied and changed by anyone.

- ☐ Multi-user
- ☐ Open source
- ☐ Proprietary

26.2 Breakin' the law!

[AQA](#)[EDEXCEL](#)[EDUQAS](#)[OCR](#)[WJEC](#)isaacss.org/bq/iot_legal_g02

Jane breaches Data Protection legislation by leaving her laptop logged on when she goes for a break at work. Amilyn uses Jane's laptop to access financial files she does not have permission to access.

► Select the piece of legislation that Amilyn has breached.

- ☐ Telecommunications Regulations Act
- ☐ Data Protection Act
- ☐ Freedom of Information Act
- ☐ Computer Misuse Act

26.3 Fact or impact? (Software licensing)

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/iot_legal_g04

Extended (long-form) questions require you to include knowledge **and** impact in your responses. This allows you to demonstrate that you can think about a concept more deeply than by only writing down points of knowledge that you can recall.

► For each sentence in the response, indicate whether the sentence demonstrates **knowledge** (e.g. of a fact/concept), or the **impact** of a fact/concept:

QUESTION: Discuss the impact of licensing on the spread of digital content.

RESPONSE: Some digital content creators use Creative Commons (CC) licences _____. CC licences allow creators to make it clear that their work can be shared and adapted, but also ensure that their work is always attributed _____. This can result in creators having their work seen by a wider audience and allows others to legally develop solutions within a community _____. The least restrictive Creative Commons licence is the attribution licence _____. When a no-derivatives licence is used it means there is no permission to modify the content _____.

☐ Knowledge

☐ Impact

26.4 Fact or impact? (Copyright)

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/iot_legal_g05

Extended (long-form) questions require you to include knowledge **and** impact in your responses. This allows you to demonstrate that you can think about a concept more deeply than by only writing down points of knowledge that you can recall.

► For each sentence in the response, indicate whether the sentence demonstrates **knowledge** (e.g. of a fact/concept), or the **impact** of a fact/concept:

QUESTION: Discuss the impact of Copyright legislation on the spread of digital content.

RESPONSE: Copyright legislation protects the rights of creators and distributors of digital content, including software _____. The legislation acts as a deterrent to people who would create copies of software without paying for it and means that creators are more likely to be remunerated for their work _____. The law was enacted in 1988 in the UK _____. Without this legal protection, there may be fewer independent developers who can focus on creating software, as they would need to find other, more financially secure employment _____. This in turn might mean that creativity and innovation could be restricted, causing less economic growth _____.

☐ Knowledge

☐ Impact

26.5 Fact or impact? (Computer Misuse Act)

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/iot_legal_g06

Extended (long-form) questions require you to include knowledge **and** impact in your responses. This allows you to demonstrate that you can think about a concept more deeply than by only writing down points of knowledge that you can recall.

► For each sentence in the response, indicate whether the sentence demonstrates **knowledge** (e.g. of a fact/concept), or the **impact** of a fact/concept:

QUESTION: Discuss the effectiveness of the Computer Misuse Act in dealing with cybercrime.

RESPONSE: The Computer Misuse Act is a piece of legislation

_____. The sections of the Act are designed to deal with a wide range of security threats _____. The sections of the Act allow the police and prosecutors to take action against cybercriminals _____. Passwords can be used to commit further offences _____. The creation and spread of malware would violate Section 3, as the program modifies the system it is installed on _____. In this way, the Act provides a clear basis to convict those guilty of a variety of cybercrimes _____. Organisations are under pressure to clearly warn potential offenders that they are not authorised to commit certain actions and the consequences of those actions _____.

- ☐ Knowledge
- ☐ Impact

26.6 Impact of legislation

AQA EDEXCEL EDUQAS OCR WJEC

isaacss.org/bq/iot_legal_g03

► Complete the table by adding the correct piece of legislation to match the relevant impact. Items can be selected more than once.

Impact	Legislation
This gives the public the right to know what public organisations spend their money on.	
Actions that employers take using this legislation can lead to the erosion of trust between employers and employees.	
Public debate can be better informed and more productive because of this.	

- ☐ Computer Misuse Act
- ☐ Copyright, Design and Patents Act
- ☐ General Data Protection Regulations (GDPR)
- ☐ Telecommunications Regulations Act
- ☐ Data Protection Act
- ☐ Freedom of Information Act

27 Impacts of technology

Technology impacts our lives in both positive and negative ways. Information is easily accessible and can be exchanged on an almost immediate basis. This has changed how much personal information is shared. It also means that it is important to be considerate of all users' needs, for example when choosing colours for a company website. Identifying impacts allows us to make better decisions as we develop and use new technologies. Awareness of impact can help avoid introducing personal biases into systems, which could affect users.

We must consider how our use of technology impacts the environment. As an example, artificial intelligence (AI) can help us identify areas vulnerable to deforestation. However, training and maintaining AI algorithms requires a lot of power, which could have a negative environmental impact.

27.1 Machine learning

AQA

EDEXCEL

OCR

isaacss.org/bq/iot_impacts_g01

► Which one of these can lead to ethical issues when making use of machine learning?

- ☐ Syntax errors
- ☐ Disposal of digital electronic equipment
- ☐ Algorithmic bias

27.2 Artificial intelligence

AQA

EDEXCEL

OCR

isaacss.org/bq/iot_impacts_g02

► What one of these is an example of an application of artificial intelligence?

- ☐ A pedestrian crossing
- ☐ Speed cameras
- ☐ Facial recognition
- ☐ A pilot flying a helicopter

27.3 Impacts

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/iot_impacts_g03

► Complete the table by adding the correct type of impact for each example.

Example	Type of impact
Social media has enabled and encouraged people from around the world to communicate with each other. The technology has led to the introduction of many new words and phrases, such as ‘emoji’, ‘selfie’, and ‘trolling’, which have now become widely used in our day-to-day conversations, as well as on social media.	
In the workplace, it is reasonable for an employee to be able to use their phone to check their social media account. It is also reasonable for an employer to monitor an employee’s use of the internet.	
Social media allows a wide range of voices to engage in public debate. This allows us to be better informed on matters that affect us.	

- ☐ Social
- ☐ Environmental
- ☐ Cultural
- ☐ Economic
- ☐ Moral
- ☐ Ethical

28 Programming concepts

Almost all programs consist of the same basic ‘building blocks’, assembled in different ways to achieve a particular goal. **Variables**, **data types**, **sequence**, **selection**, and **iteration** are examples of these basic concepts, which all new programmers need to learn. In order to compare and manipulate data, and to make decisions in a program, programmers use a variety of different **logical**, **relational**, and **arithmetic operators**. It is also useful to be able to generate **randomness** in a program, for example selecting a random number or a random item from a list.

28.1 Odd or even?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_prog_05

Martin wrote a program a few weeks ago, but he has forgotten to comment his code, so he can’t remember how it works.

```
1  end = 10
2  FOR i = 1 TO end
3      IF (i MOD 2) == 0 THEN
4          PRINT("red")
5      ELSE
6          PRINT("blue")
7      ENDIF
8  NEXT i
```

► What will be printed when `i` reaches 8 in the loop?

- ☐ blue
- ☐ red
- ☐ 0
- ☐ 8

28.2 List the coordinates

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_prog_02

Sian is writing a program for a game with a grid. Here is the grid she is using:

x \ y	1	2	3
1			
2		@	
3			
4	#		

Sian wants to write some code to generate the grid positions of every shaded square on her grid so that she can check them. For example, the co-ordinates of the square containing the @ sign would be $x=2$ $y=2$.

► Which of these pieces of code generates all of the shaded grid positions? Circle the correct answer.

OPTION A

```

1  x = 1
2  y = 1
3  WHILE x <= 4 AND y <= 3
4      PRINT("x=" + STR(x) + " y=" + STR(y))
5      x = x + 1
6      y = y + 1
7  ENDWHILE

```

OPTION B

```

1  FOR x = 1 TO 4
2      FOR y = 1 TO 3
3          PRINT("x=" + STR(x) + " y=" + STR(y))
4      NEXT y
5  NEXT x

```

OPTION C

```

1  FOR x = 1 TO 7
2      PRINT("x=" + STR(x) + " y=" + STR(x - 1))
3  NEXT x

```

OPTION D

```

1  FOR x = 1 TO 3
2      FOR y = 1 TO 4
3          PRINT("x=" + STR(x) + " y=" + STR(y))
4      NEXT y
5  NEXT x

```

28.3 Which code will execute?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_prog_01

```
1 number = 5
2 triple = True
3
4 IF triple == True AND number >= 10 THEN
5     number = number * 3
6 ELSE
7     number = number + 5
8 ENDIF
```

► What is the value of the variable `number` once the code has finished executing?

28.4 Constants

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_prog_08

A banking system uses a constant to store the current rate of VAT so that it can be used in calculations.

```
1 VAT_RATE = 0.20
```

► Which of the following statements relating to constants is true?

- ☐ A constant is the same as a variable.
- ☐ The value of a constant can only change while the program is running.
- ☐ You cannot change the value of a constant in the program source code.
- ☐ A constant stores a value that does not change during the execution of the program.

28.5 Not in stock?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_prog_11

A fruit seller has created a program, shown in pseudocode below, to help their customers check the level of fruit in stock.

```
1  PROCEDURE check_stock()
2      fruit = INPUT("What fruit would you like ?")
3      IF fruit == "Mango" THEN
4          level = "low"
5      ELSEIF fruit != "Banana" AND fruit != "Apple" THEN
6          level = "no"
7      ELSE
8          level = "good"
9      ENDIF
10     PRINT("There is " + level + " stock")
11 ENDPROCEDURE
```

► From the list of fruits that follows, select all the fruits which have a **low** or **good level of stock**.

- ☐ Peach
- ☐ Banana
- ☐ Apple
- ☐ Mango
- ☐ Orange

28.6 Modulo

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/program_concepts_arithmetic_operators_g02

► What would be the output from the following program?

```
1  result = 20 MOD 8
2  PRINT(result)
```

28.7 A trace table

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/gcse_prog_14

A student has designed an algorithm that is shown in pseudocode below. The algorithm uses a condition-controlled loop.

```
1 PROCEDURE main()
2     input_value = INPUT("Enter a number: ")
3     num = INT(input_value)
4     count = 0
5     WHILE num < 500
6         num = num * 2
7         count = count + 1
8     ENDWHILE
9     PRINT(count - 1)
10 ENDPROCEDURE
```

The student traces the algorithm for a user input of 16.

► Which of the following options shows a trace table with the correct values?

○

Trace table			
num	count	num < 500	OUTPUT
16	1	TRUE	
32	2	TRUE	
64	3	TRUE	
128	4	TRUE	
256	5	TRUE	
512	6	FALSE	
			5

○

Trace table			
num	count	num < 500	OUTPUT
16	0	TRUE	
32	1	TRUE	
64	2	TRUE	
128	3	TRUE	
256	4	TRUE	
512	5	FALSE	
			4

☐

Trace table			
num	count	num < 500	OUTPUT
16	0	TRUE	
32	1	TRUE	
64	2	TRUE	
128	3	TRUE	
256	4	TRUE	
		FALSE	3

☐

Trace table			
num	count	num < 500	OUTPUT
16	0	TRUE	
32	1	TRUE	
64	2	TRUE	
128	3	TRUE	
256	4	TRUE	
512	5	FALSE	
			511

28.8 Spot the data type

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_concepts_data_types_g01

Take a look at the data presented in the options below.

► Select the option that is most likely to be an example of the **string** data type.

- ☐ True
- ☐ Five
- ☐ 5.1
- ☐ 5

28.9 Spot the data type 2

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_concepts_data_types_g02

Take a look at the data presented in the options below.

► Select the option that is most likely to be an example of the **integer** data type.

- ☐ 5.1
- ☐ False
- ☐ Five
- ☐ 5

28.10 Integer division

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/program_concepts_arithmetic_operators_g01

► What would be the output from the following program?

```
1 result = 20 DIV 8
2 PRINT(result)
```

28.11 Predict the output

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/program_concepts_data_types_g03

► What will be the output of line 5 when the code is executed if the user enters 1 for the first number and 7 for the second number?

```
1 PRINT("Enter a number")
2 num1 = INPUT()
3 PRINT("Enter another number")
4 num2 = INPUT()
5 PRINT(num1 + num2)
```

☐ 8☐ 17

28.12 Accepting input in Python

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/program_concepts_inputs_g01

► Which of the following instructions are carried out by an `input` function. Choose **three** correct answers from the list of five possible answers.

- ☐ It takes the value entered by the user
- ☐ It returns the converted value as string
- ☐ It checks the data type of the entered value
- ☐ It returns the value using the specified data type
- ☐ It converts the value into the string data type

28.13 Converting data types

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/prog_concepts_inputs_g02

► Which line of code in the program below contains a function that will convert a string to an integer?

```
1 PRINT("How old are you?")
2 user_input = INPUT()
3 age = INT(user_input)
4 months = age * 12
5 PRINT("Wow, you have been alive for around " + STR(months) +
    " months!")
```

- ☐ PRINT("Wow, you have been alive for around " + STR(months)+ " months!")
- ☐ age = INT(user_input)
- ☐ PRINT("How old are you?")
- ☐ months = age * 12
- ☐ user_input = INPUT()

28.14 User input

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/prog_concepts_inputs_g03

► The code below is executed. If the user enters the string **zero** when prompted for a decimal number, what will happen?

```
1 PRINT("Enter a decimal number")
2 user_input = INPUT()
3 num1 = FLOAT(user_input)
4 PRINT("Your decimal number is: " + STR(num1))
```

- ☐ It will remind the user to enter a number and prompt them to enter it again
- ☐ It will display Your decimal number is: zero
- ☐ A value error will occur
- ☐ It will convert the string **zero** to the decimal number 0.0

28.15 Step in time

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/programming_concepts_iteration_g01

► What will be the output of the following program when it is executed?

```
1 FOR i = 1 TO 9 STEP 3
2     PRINT(i)
3 NEXT i
```

- ☐ 1,2,3
- ☐ 4,7
- ☐ 1,2,4,5,6,7,8,9
- ☐ 1,4,7

28.16 Top of the league

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/programming_concepts_iteration_g02

Samir has created a program that keeps track of his team's score for a local table tennis league. Below is the program that he has written.

```
1 red_team = ["W","W","L","D"]
2 win = 4
3 draw = 2
4 loss = -1
5 total = 0
6
7 FOR EACH result IN red_team
8     IF result == "W" THEN
9         total = total + win
10    ELSEIF result == "D" THEN
11        total = total + draw
12    ELSE
13        total = total + loss
14    ENDIF
15 NEXT result
16 PRINT(total)
```

► What will be the output of his program when it is executed?

- ☐ 2
- ☐ 4
- ☐ 11
- ☐ 9

28.17 Countdown

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_concepts_iteration_g03

► What will be the output of this program when it is executed?

```
1 timer = 3
2 WHILE timer > 0
3     PRINT(timer)
4     timer = timer - 1
5 ENDWHILE
6 PRINT("Time is up")
```

- ☐ 3,Time is up, 2,Time is up, 1,Time is up,
- ☐ 3,2,1,Time is up
- ☐ 3,2,1,0,Time is up
- ☐ Time is up

28.18 Crazy weather

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_concepts_iteration_g04

► What will be the output of this program when it is executed?

```
1 weather = "Sun"
2
3 FOR counter = 1 TO 3
4     PRINT(weather)
5     weather = "Rain"
6 NEXT counter
7
8 IF weather == "Rain" THEN
9     PRINT("It's raining")
10 ENDIF
```

- ☐ Sun, It's raining
- ☐ Sun, Rain, Rain
- ☐ Sun, Rain, Rain, It's raining
- ☐ Sun, Rain, It's raining

28.19 Random ints

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/prog_concepts_random_numbers_g01

A learner wants to use a program to generate random numbers for a dice rolling game. The code can be seen below:

```
1 dice_roll = RANDOM_INT(1,7)
2 PRINT(dice_roll)
```

► What will happen when this program is executed? Choose **one** from the list of four options.

- ☐ Either 1, 2, 3, 4, 5, or 6 will be displayed as output on the screen
- ☐ Either 1, 2, 3, 4, 5, 6, or 7 will be displayed as output on the screen
- ☐ RANDOM_INT(1,7) will be displayed on screen
- ☐ dice_roll will be displayed on the screen

28.20 IF/ELSE

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/prog_concepts_selection_g02

► What will be the output of the following program when it is executed?

```
1 number = 15
2 IF number > 13 THEN
3     PRINT(number + 3)
4 ELSE
5     PRINT(number + 10)
6 ENDIF
```

- ☐ 25
- ☐ 18
- ☐ 18, 25
- ☐ 16

28.21 What IF you don't get enough sleep?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_concepts_selection_g01

Rose has designed an application that promotes healthy lifestyle choices. She would like the app to do the following:

- Allow the user to input the number of hours they slept
- Provide a message to the user based on the number of hours that they slept

► Put the blocks of code in the correct order to form a working segment of code.

- ☐ PRINT("Well done. 8+ hours is perfect! Keep up the good work")
- ☐ PRINT("6+ hours. Not bad. It is recommended that you get 6 hours sleep as a minimum")
- ☐ hours_slept = INT(INPUT())
- ☐ ELSE
- ☐ IF hours_slept >= 8 THEN
- ☐ PRINT("You should aim to get between 6 and 8 hours sleep")
- ☐ ENDIF
- ☐ ELSEIF hours_slept >= 6 THEN

28.22 Is it morning?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/program_concepts_selection_g03

► What will be the output of the following program when it is executed?

```
1 hour = 12
2 IF hour <= 12 THEN
3     PRINT("Good morning")
4 ENDIF
5
6 PRINT("How are you?")
```

- ☐ Good morning, How are you
- ☐ Good morning
- ☐ How are you?

28.23 Greater or less than?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/program_concepts_selection_g04

► What will be the output of the following program when it is executed?

```
1 number_1 = 30
2 number_2 = 90
3
4 IF number_1 > 30 AND number_2 == 90 THEN
5     PRINT("1")
6 ELSEIF number_1 > 30 OR number_2 == 90 THEN
7     PRINT("2")
8 ELSEIF number_1 == 30 AND number_2 == 90 THEN
9     PRINT("3")
10 ELSE
11     PRINT("4")
12 ENDIF
```

- ☐ 1
- ☐ 4
- ☐ 2
- ☐ 2,3

28.24 Variable assignment 1

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/prog_concepts_variables_g01

► What will be the output of PRINT when this program is executed?

```
1 PRINT(player_name)
2 player_name = "BlueSky93"
```

- ☐ There is an error in the program
- ☐ player_name
- ☐ BlueSky93

28.25 Variable assignment 2

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/prog_concepts_variables_g02

► What will be the output of PRINT when this program is executed?

```
1 player_name = "BlueSky93"
2 PRINT(player_name)
```

- ☐ player_name
- ☐ "BlueSky93"
- ☐ There is an error in the program
- ☐ BlueSky93

28.26 Variable assignment 3

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/prog_concepts_variables_g03

► What will be the output of PRINT when this program is executed?

```
1 player_name = "BlueSky93"
2 player_name = "DinoFish40"
3 PRINT(player_name)
```

- ☐ BlueSky93
- ☐ BlueSky93, DinoFish40
- ☐ DinoFish40
- ☐ player_name

29 String handling

A **string** is a sequence of characters (letters, numbers, spaces, or other symbols). String handling techniques are used to check and manipulate string data, for example to determine whether a password is sufficiently long, or to remove or replace characters in a piece of text.

This topic uses a single version of pseudocode and versions of real program code in Python and C#.

29.1 Number plate substrings

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/gcse_prog_03

Nitin owns a car showroom, and he has records of all of the cars he has in stock. The records include the cars' number plates. The number plates are all in the following format, where L is a letter and N is a number:

LLNNLLL

Nitin has written a program to try to extract only the two numbers (NN) from the number plate of every car, which represent the year in which it was registered.

The documentation for the programming language said, “to obtain a substring, you must specify the starting position (numbering starts at 0), the ending position (inclusive), and the string itself”.

He isn't sure whether the program he has written is correct. Here is his code:

```
numberplate = "ZX19CSV"  
year = numberplate[2:5]  
PRINT(year)
```

► What will Nitin's program print out?

29.2 Check the password

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_string_g01

A web designer is developing an application where the user has to create a password. The password has to be between 10 and 16 characters and must not contain any blank spaces.

The pseudocode of the `check_password` function below is used to validate the password created by the user.

```
1 FUNCTION check_password(password)
2     is_valid = True
3     password_length = LEN(password)
4     IF password_length < 10 OR password_length >16 THEN
5         is_valid = False
6     ELSE
7         FOREACH character IN password
8             IF character == " " THEN
9                 is_valid = False
10            NEXT character
11        ENDIF
12    RETURN is_valid
13 ENDFUNCTION
```

► Select the option that best describes why the `check_password` function returns **False** when it is called with the argument “My Great Password”?

- ☐ It returns **False** because the password length is three words
- ☐ It returns **False** because the password has two space characters
- ☐ It returns **False** because the password length is 17 characters

29.3 STRIP a substring

EDUQAS

isaac.cs.org/bq/prog_string_g02

A programmer receives a file.

Each line of text in the file has an unwanted '*' character at the start.

Here is part of the code the programmer writes to remove the unwanted character from each line.

```
1 old_file = OPEN_READ("old.txt")
2 lines = old_file.READ_ALL()
3 old_file.CLOSE()
4
5 CREATE_FILE("new.txt")
6 new_file = OPEN_APPEND("new.txt")
7
8 FOREACH line IN lines
9     cleaned = *** missing code ***
10     new_file.WRITE_LINE(cleaned)
11 NEXT line
12
13 new_file.CLOSE()
```

► Select the line of code the programmer should use to remove the unwanted character.

- ☐ STRIP "*" from line)
- ☐ STRIP "*" from EACH line)
- ☐ STRIP(line, "*")
- ☐ STRIP("*", line)

29.4 Concatenating strings

AQA

EDEXCEL

EDUQAS

OCR

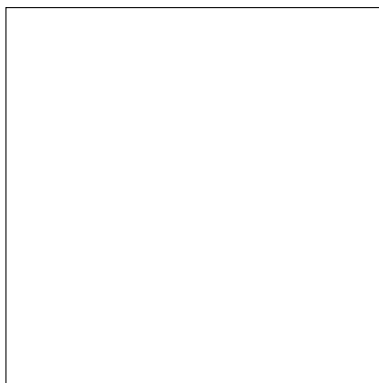
WJEC

isaacacs.org/bq/prog_string_g03

► Using **four** of the lines of code supplied, create a program that uses concatenation to produce this output:

Venus is the hottest planet.

- ☐ `string = name + text`
- ☐ `print(string)`
- ☐ `name = "Venus "`
- ☐ `text = "is the hottest planet."`
- ☐ `print(Venus is the hottest planet.)`



29.5 What's in this string?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_string_g04

► Which one of these shows a variable that has been assigned an empty string?

- ☐ `word = 0`
- ☐ `word = "0"`
- ☐ `word = " "`
- ☐ `word = ""`

29.6 Indexing

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_string_g05

A variable named `legend` is assigned a string:

```
legend = "Lovelace"
```

► What is the character returned by `legend[2]`?

☐ L☐ o☐ v

29.7 Getting the position

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_string_g06

► What is output when this code runs?

```
1 position = INDEXOF("Garlic", "i")
2 PRINT(position)
```

☐ 4☐ 5☐ 6

29.8 A slice of "Pie"

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_string_g07

► What substring is output when this code runs?

```
1 name = "Pie"
2 PRINT(name[:-1])
```

☐ Pi☐ name[:-1]☐ ie☐ P

29.9 Validation

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/prog_string_g08

► Which one of these is the reason for using `LOWER()` when validating user input to this prompt?

```
1 answer = "Would you like to continue? Enter Y or N: "
```

- ☐ To make sure the output data is correct
- ☐ To check the line below if the user enters a character on the wrong line
- ☐ To handle a string of more than one character entered by the user
- ☐ To handle a string of an unexpected case entered by the user

30 Subroutines

A **subroutine** is a named section of code that can be called by writing the subroutine's name in a program statement. Subroutines can also be referred to as **procedures** or **functions**. A **procedure** is a set of instructions that will be executed when the procedure is called, whilst a **function** also contains a set of instructions that will be executed when the function is called and it will also return a value when it has finished executing.

Using subroutines makes the code more readable and reusable, as the code is broken into smaller sections. Most programming languages come with a standard library of built-in subroutines to perform common functions, but also have the capability to allow the programmer to write their own custom subroutines.

30.1 Global variables

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_sub_g05

► From the sentences below select the one that best describes a **global variable**.

- ☐ A **global variable** refers to a variable that can only be accessed and modified within the code block in which it is declared.
- ☐ A **global variable** is declared in the main part of the program and can be accessed globally by all subroutines.
- ☐ A **global variable** is declared in the main part of the program but cannot be accessed globally by all subroutines.

30.2 Using subroutines

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_sub_g06

► Select the statement that is true.

- ☐ Subroutines make programs easier to read and understand, because they break program code into smaller sections.
- ☐ Procedures allow values to be returned to the procedure call.
- ☐ If a subroutine is defined, then it must be executed at some point in the program.

30.3 Fix the calculator

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_prog_10

George is writing a calculator program. The code he has written so far, is shown in the following pseudocode:

```
1  PROCEDURE multiply(value_a, value_b)
2      PRINT(value_a * value_b)
3  ENDPROCEDURE
4
5  PROCEDURE main()
6      first = INPUT("Enter the first number: ")
7      first = INT(first)
8      second = INPUT("Enter the second number: ")
9      second = INT(second)
10     result = multiply(first, second)
11     PRINT("The result is " + result)
12 ENDPROCEDURE
```

George tests his program by typing in the numbers 5 and 2, and this is what he sees:

```
1  Enter the first number: 5
2  Enter the second number: 2
3  10
4  The result is null
```

► Why does the final line state that the result is null?

- ☐ He didn't call the subroutine correctly.
- ☐ He forgot to save the return value from the subroutine into a variable.
- ☐ He has not specified a return value from the subroutine.
- ☐ The print statement on line 2 caused an error.

30.4 Return a value

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/gcse_prog_13

The cube of a given number is that number multiplied by itself three times. For example, the cube of 3 is 27.

Lucy has created a program (shown in pseudocode below) that uses a subroutine named `cubed` to carry out the calculation. When she runs the program, Lucy is not getting the output that she was expecting.

```
1 FUNCTION cube(num)
2     cubed_value = num * num * num
3     RETURN cubed_value
4 ENDFUNCTION
5
6 PROCEDURE main()
7     input_value = INPUT("Enter a value to be cubed: ")
8     user_num = INT(input_value)
9     cube(user_num)
10 ENDPROCEDURE
```

Lucy realises that problem is with the final line of code.

► How should she amend it so that the cubed value is outputted correctly?

- ☐ PRINT(cube(user_num))
- ☐ PRINT(cubed_value)
- ☐ PRINT(num)
- ☐ PRINT(user_num)

30.5 Parameters

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_sub_g01

Parameters are defined when a subroutine is _____, usually inside brackets after the name of the subroutine (the subroutine _____).

The parameters are also given identifiers (names). A subroutine can have one, more than one, or _____ parameters.

► Use the words below to complete the above paragraph.

- ☐ no
- ☐ identifier
- ☐ called
- ☐ declared

30.6 Arguments

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_sub_g02

An argument is any item of data that is passed to a _____.

The subroutine must be called with the number of arguments defined by its list of _____.

► Use the words below to complete the above paragraph.

- ☐ subroutine
- ☐ parameters
- ☐ items
- ☐ program

30.7 Procedures

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_sub_g03

A student wants a subroutine that will add any two given numbers together and to display the answer.

► Use the lines of code below to construct a `display_sum` procedure for them.

- ☐ `ENDPROCEDURE`
- ☐ `PRINT (answer)`
- ☐ `answer = num1 + num2`
- ☐ `PROCEDURE`
 `display_sum(num1,`
 `num2)`

30.8 Passing data

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_sub_g08

► Rearrange the lines of code below to construct a function, then call the function and pass data to it.

- ☐ `FUNCTION`
 `calculate(num1, num2)`
- ☐ `ENDFUNCTION`
- ☐ `PRINT (answer)`
- ☐ `answer = calculate(5,`
 `10)`
- ☐ `RETURN answer`

30.9 Functions

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/prog_sub_g09

A _____ is a named block of code that has the purpose of _____ a value. A _____ can be called by another part of the program but does not return a value.

► Use the words below to complete the paragraph above.

- ☐ returning
- ☐ value
- ☐ procedure
- ☐ function

30.10 Local variables

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/prog_sub_g04

► From the sentences below select the one that best describes a **local variable**.

- ☐ **Local variables** are variables that cannot be accessed and modified within the code block in which they were declared.
- ☐ **Local variables** are variables that can only be accessed and modified within the code block in which they were declared.
- ☐ **Local variables** are variables that can be accessed and modified from anywhere in the program.

30.11 Advantages of subroutines

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_sub_g07

► Which one of the following statements describes an advantage of subroutines?

- ☐ When creating very large programs, you can give different programmers different subroutines to write, and these can then be combined.
- ☐ Subroutines are always defined by the user, and these cannot be combined.
- ☐ Procedures allow values to be returned to the procedure call.

30.12 Multiplication machine

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_sub_g10

You have been tasked with creating a program that multiplies any two numbers together.

► Use the options below to complete the following program:

```

1  ----- multiply_two_numbers(number1, number2)
2      answer = number1 * number2
3  ----- answer
4  -----
5
6  result = multiply_two_numbers(5,5)
7  PRINT(result)

```

☐ ENDFUNCTION☐ PROCEDURE☐ ENDPROCEDURE☐ PRINT☐ RETURN☐ FUNCTION

30.13 Split the bill

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/gcse_prog_09

Maja has written a program that is supposed to display how much one person should pay if a restaurant bill is split equally between the people present.

The program isn't working correctly; when Maja runs it, the amount does not appear on the screen.

```
1 FUNCTION split_the_bill(amount, number_of_people)
2     result = FLOAT(amount) / INT(number_of_people)
3     RETURN result
4 ENDFUNCTION
5
6 PROCEDURE main()
7     bill_total = INPUT("Enter the total bill amount")
8     people = INPUT("How many people were eating?")
9     PRINT("The amount you should pay is...")
10    split_the_bill(bill_total, people)
11 ENDPROCEDURE
```

► Why doesn't the program print the amount?

- ☐ She has got the bill calculation wrong.
- ☐ She has called the subroutine correctly, but has not displayed the return value.
- ☐ She has forgotten to call the subroutine.
- ☐ She has put the arguments the wrong way round when calling the subroutine.

31 File handling

Many computer applications use some form of persistent data. This is data that is stored in secondary storage and will be available whenever the application is used. All persistent data is stored in files. This can be a single file, or for larger systems a collection of related files. Some applications use databases. Databases are large complex file systems where all of the low level file organisation and access is managed for you. However, databases are not suitable for every application; sometimes a more simple file structure is needed.

31.1 File mode

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/prog_files_g01

Natalia is a software developer for a clothes retailer. She has a csv file named ‘stock.csv’ that includes information on all of the clothing sold by the retailer.

Task	Mode
Add new new items of clothes to ‘stock.csv’	
Search the file for a specific item of clothes	
Create a new file named ‘search-results.csv’.	
Overwrite contents of ‘search-results.csv’ with the results of each new search	

► Look through the list of jobs above Natalia has to perform and decide which file handling mode below would be most suitable for each task.

- ☐ Write
- ☐ Append
- ☐ Read

31.2 CSV Downloads

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/prog_files_g03

Below are the contents of a file named *user_download_history.csv* showing app downloads for a customer.

DownloadID	AppTitle	PurchaseDate	Price
187362	Monopoly Classic	2022-03-01	4.99
273340	Mini Football	2022-03-20	2.79
317333	Stardew Valley	2021-11-19	4.99
387737	Game Dev Tycoon	2021-04-28	4.99
622100	Alien Isolation	2022-02-12	12.99

The user wants a list of only the titles of all the apps they have purchased.
► Use the items below to complete the subroutine that will print a list of titles for the customer. The field that contains the title is **AppTitle**.

```
1 SUBROUTINE read_csv()
2     file_object =
3         _____('user_download_history.csv')
4     WHILE NOT file_object.END_OF_FILE()
5         record = file_object.READ_LINE()
6         app_data = _____(record,
7             '_____')
8         app_title = movie_data[_____]
9
10        PRINT('App title: ' + _____)
11    ENDWHILE
12    file_object.CLOSE()
13 ENDSUBROUTINE
```

- ☐ OPEN_READ
- ☐ 1
- ☐ ,
- ☐ 0
- ☐ SPLIT
- ☐ app_title
- ☐ AppTitle
- ☐ SEPARATE

31.3 What's in the file?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_files_g02

Kasim has a file with the name `test_data.csv`. He needs to write code to be able to open the file and to output the first line so that knows the headings of each column.

► Reorder the code below so that Kasim has a working program that reads and outputs only the first line of the file.

Note: Not all lines of code provided are needed to complete the program

- ☐ `file_object = OPEN_WRITE("bridge.txt")`
- ☐ `PRINT(headings)`
- ☐ `headings = file_object.READ_LINE()`
- ☐ `file_object.CLOSE()`
- ☐ `file_object = OPEN_READ("bridge.txt")`

31.4 Adding students to the list

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_files_g04

A teacher uses this code to display a list of students' names.

```

1 PROCEDURE read_names()
2     file_object = OPEN_READ('list.txt')
3     list = file_object.READ_ALL()
4     file_object.CLOSE()
5     PRINT(list)
6 ENDPROCEDURE

```

New code is needed to add new students' names to the list.

► Using the code options provided, add the lines of code into the correct order to complete the new procedure. You do not have to use all the lines of code provided.

```

1 PROCEDURE add_name(new_name)
2     -----
3     -----
4     -----
5 ENDPROCEDURE

```

☐ file_object = OPEN_APPEND("list.txt")

☐ file_object.WRITE_LINE(new_name)

☐ file_object.CLOSE()

☐ file_object = OPEN_READ("list.txt")

☐ file_object.READ_LINE()

☐ file_object = OPEN_WRITE("list.txt")

31.5 Missing movie

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_files_g06

A file called `movies.csv` contains a list of movies.

This code has been developed to add details of new movies to the file.

► Add the code into the spaces to ensure the code successfully adds new movies to the file. You do not have to use all the lines of code provided.

```

1  PROCEDURE main()
2      title = INPUT('Enter the movie title: ')
3      year = INPUT('Enter the movie year: ')
4      duration = INPUT('Enter the length of the movie (01:00):
        ')
5      director = INPUT('Enter the name of the movie's
        director: ')
6      rating = INPUT('Enter the average movie rating (1 to 5):
        ')
7      price = INPUT('Enter the price in pounds (£00.00):')
8      append_movie_optimised(title, year, duration, director,
        rating, price)
9  ENDPROCEDURE
10
11 PROCEDURE _____(title, year, duration, director,
    rating, price)
12     _____ = title + ',' + year + ',' + duration
        + ',' + director + ',' + rating + ',' + price
13     file_object = _____('movies.csv')
14     _____(new_movie)
15     file_object._____()
16 ENDPROCEDURE

```

☐ `append_movie_optimised`
☐ `ENDPROCEDURE`
☐ `OPEN_WRITE`
☐ `CLOSE`
☐ `OPEN_APPEND`
☐ `file_object.WRITE_LINE`
☐ `new_movie`
☐ `OPEN_READ`

31.6 Outputting the contents of a file

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/prog_files_g05

► Which one of these options should be used to open a file to output its contents?

- ☐ `OPEN.APPEND('food.txt')`
- ☐ `OPEN.WRITE('food.txt')`
- ☐ `OPEN('food.txt')`
- ☐ `OPEN.READ('food.txt')`

31.7 Is it true?

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/prog_files_g08

► Select the **three** correct statements about file handling.

- ☐ The append mode adds new data at the end of the file.
- ☐ The use of write mode can overwrite the existing data of a file.
- ☐ The append and write modes both allow a file to be read.
- ☐ The read mode allows new data to be added to the end of the file.
- ☐ Only read mode allows the file to be read.

31.8 End of the line!

AQA EDEXCEL EDUQAS OCR WJEC

isaacacs.org/bq/prog_files_g10

► The file `teams.csv` contains the details of ten sports teams.

Which one of these options is a possible output from the code `teams.END_OF_FILE()`?

- ☐ `TRUE`
- ☐ `.csv`
- ☐ The last line of data

31.9 The charity shop system

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/prog_files_g07

► A charity shop needs a stock management system. The system should allow staff members to add new items and see all current stock. The code contains some gaps. Add the code into the spaces, so the code successfully adds a new item of stock and can print a list of all stock.

```

1  PROCEDURE add_items():
2      code = INPUT('Enter Item Code')
3      name = INPUT('Enter Item Name')
4      price = INPUT('Enter Item Price')
5      file = _____('items_list.txt')
6      file.WRITE_LINE(code + ' - ' + name + ' - ' + price)
7      file._____()
8  ENDPROCEDURE
9
10 PROCEDURE disp_items():
11     file = OPEN_READ('_____')
12     items = file._____()
13     FOREACH item IN items:
14         PRINT(item)
15 END_____
16
17 PROCEDURE read_menu():
18     menu = '1'
19     WHILE menu != '3':
20         menu = INPUT('1 - Add Item \n 2 - Display Item \n 3 -
21             Exit \n')
22         IF menu == '1':
23             _____()
24             ELIF menu == '2':
25                 disp_items()
26             ELSE:
27                 PRINT('Goodbye!')
28 ENDPROCEDURE

```

☐ CLOSE☐ PROCEDURE☐ OPEN_APPEND☐ add_items☐ READ_LINE☐ items_list.txt

31.10 File writing

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_files_g09

► A program should write a new customer's details to a file called `customers.txt`. A procedure has been started. Add the lines of code provided into the correct order.

```
1  PROCEDURE add_new_customer(firstname, surname, address,
    tel_number)
2  -----
3  -----
4  -----
5  ENDPROCEDURE
```

☐ `file = OPEN_APPEND("customers.txt")`☐ `file.WRITE_LINE(firstname, surname, address, tel_number)`☐ `file_object.CLOSE()`

32 Integrated development environments (IDEs)

An **integrated development environment (IDE)** is software that includes various features to help a programmer develop their program code. Most IDEs will include facilities such as automatic formatting and debugging tools such as break points. It is common for an IDE to allow code to be executed from inside the environment, with the help of an **interpreter** or **compiler**.

32.1 What is an IDE?

EDUQAS OCR WJEC

isaacacs.org/bq/prog_softeng_ide_g01

Most programmers use an integrated development environment (IDE) when working on program code.

- Which of the following statements most accurately describes an IDE?
- ☐ Software that is designed with the sole purpose of debugging your program.
 - ☐ Software that supports you to write code, to execute your program, and to debug errors.
 - ☐ Software that includes a text editor but requires you to use additional software with a runtime environment to execute the code.

32.2 Debugging

EDUQAS WJEC

isaacacs.org/bq/prog_softeng_ide_g03

► Which two of the following software features commonly included with an integrated development environment (IDE) would be useful when debugging a logic error in your code?

- ☐ Break point
- ☐ Automatic formatting
- ☐ Syntax checking
- ☐ Threading

32.3 IDE features

EDUQAS OCR WJEC

isaacss.org/bq/prog_softeng_ide_g02

The table below provides definitions for some of the features of an integrated development environment (IDE).

► Put the words into the spaces provided to match the feature to the definition.

Definition	Feature
The window in your IDE that allows you to write your code	
Allows you to add a pause to your program to stop the flow of execution, and step through the following code line by line to trace the values of the variables	
Colour that is applied to key elements of code, such as variables, functions, and data type	
Uses a translator to execute your program code	
Connects pre-compiled libraries to your program so that they can be used together	
A feature that spots problems in your code as you write it, which would cause an error when the program is executed	

- ☐ Linker
- ☐ Text editor
- ☐ Keyword highlighting
- ☐ Break point
- ☐ Syntax checker
- ☐ Runtime environment

32.4 Benefits of an IDE

EDUQAS

OCR

WJEC

isaacacs.org/bq/prog_ide_g02

► The following description highlights the features and benefits of using an IDE. Fill in the gaps using the word provided below.

_____ helps to _____ your _____ by reducing typos and helping you to avoid common mistakes, such as forgetting to close a bracket. Automatic formatting can improve efficiency because it makes sure your code matches the requirements of the programming language. Languages such as Python strictly require code to be indented to indicate the program _____.

Debugging tools make it easier to track down and fix errors in programs. A break point adds a pause to the program to stop the flow of execution. This allows you to _____ through the code up to that point, rather than the _____ program. A memory inspector helps to debug problems with memory allocation in the program. It can help you to identify problems such as a memory leak: the program doesn't release _____ from the memory when it is no longer in use.

- ☐ flow
- ☐ step
- ☐ Autocomplete
- ☐ productivity
- ☐ whole
- ☐ speed up
- ☐ data

32.5 Loaders

EDUQAS

WJEC

isaaccs.org/bq/prog_ide_g01

► Which **one** of the following functionalities is carried out by a **loader** in an IDE?

- ☐ Provides libraries into the memory in preparation for the program to be executed.
- ☐ Displays the contents of memory so that you can see how it is being used.
- ☐ Connects libraries to the program so that they can be used together.

33 Procedural programming

The term **programming paradigm** is used to specify an overall approach to writing program code. Procedural programming is a programming paradigm that has its roots in **structured programming**.

In procedural programming, programs are made up of **procedures**, also known as subroutines or functions. Each procedure contains a series of computational steps that will be carried out when the procedure is called.

Some programming languages support only the procedural paradigm. Others are purely object-oriented or functional. If you have used Python, a popular programming language that supports all three of the paradigms mentioned, you will probably have started your programming journey by taking a procedural approach to program development.

33.1 Find the identifier

AQA

isaacacs.org/bq/prog_pas_g01

The following subroutine, expressed in pseudocode, forms part of a structured program.

```
1 FUNCTION find_highest_number(number1, number2)
2     result = ""
3     IF number1 > number2 THEN
4         result = "number 1 is highest"
5     ELSEIF number2 > number1 THEN
6         result = "number 2 is highest"
7     ELSE
8         result = "both numbers are equal"
9     ENDIF
10    RETURN result
11 ENDFUNCTION
```

► What is the identifier of the subroutine?

► Give the identifier of a parameter for the subroutine shown.

33.2 Structured benefits

AQA

isaacccs.org/bq/prog_pas_g02

► The structured approach to programming provides many benefits. From the list below, select all of the statements that specify a benefit of using the structured approach. From the list below select all of the statements that specify a benefit of using the structured approach.

- ☐ A structured program will execute more quickly than an unstructured program.
- ☐ Meaningful identifiers for subroutines help to make the program code self-documenting.
- ☐ It is quicker to develop applications because many subroutines can be worked on at the same time by different members of the development team.
- ☐ Subroutines make use of parameters and return values to interface with each other so that the use of global variables is reduced or eliminated.
- ☐ Subroutines are easier to test because they perform a single task.

33.3 Find the exit

AQA

isaacccs.org/bq/prog_pas_g03

The following function, expressed in pseudocode, converts an error code into an error message.

```
1 FUNCTION convert_code(error_code)
2     IF error_code == "400" THEN
3         RETURN "Bad request"
4     ELSEIF error_code == "404" THEN
5         RETURN "Not found"
6     ELSEIF error_code == "408" THEN
7         RETURN "Request Time-Out"
8     ELSE
9         RETURN "Error!"
10    ENDIF
11 ENDFUNCTION
```

► How many 'exit points' does the function have?

34 Object-oriented programming

Object-oriented programming (OOP) is an approach to systems development that focuses on objects and the way that they interact. Objects surround us in the real world. Look around you. For example, I am a person, and I have a desk, a computer, and a cup of water; these are all objects. How do they interact? I sit at my desk, switch on the computer, and drink some water.

The OOP method of programming (or **paradigm**) is different to the procedural paradigm, which many programmers start out with on their learning journey. In this topic you will learn about objects and how they can be defined and worked with in a computer system.

34.1 Radio one

EDUQAS

WJEC

isaacss.org/bq/prog_oop_g01

The following class has been defined using pseudocode.

```
1  CLASS Radio
2      PRIVATE volume: integer
3      PRIVATE station: string
4      PRIVATE on: Boolean
5
6      PUBLIC PROCEDURE Radio(given_station)
7          station = given_station
8          volume = 3
9          on = False
10     ENDPROCEDURE
11
12     PUBLIC FUNCTION get_volume()
13         RETURN volume
14     ENDFUNCTION
15
16     PUBLIC FUNCTION get_station()
17         RETURN station
18     ENDFUNCTION
19
20     PUBLIC FUNCTION is_on()
21         RETURN on
22     ENDFUNCTION
23
24     PUBLIC PROCEDURE set_volume(new_volume)
25         volume = new_volume
```

```

26      ENDPROCEDURE
27
28      PUBLIC PROCEDURE set_station(new_station)
29          station = new_station
30      ENDPROCEDURE
31
32      PUBLIC PROCEDURE switch()
33          IF on == True THEN
34              on = False
35          ELSE
36              on = True
37          ENDIF
38      ENDPROCEDURE
39  ENDCLASS

```

► Which of the following options (expressed in pseudocode) will correctly create an instance of a radio object?

- ☐ Radio = NEW radio1("Capital FM", 3, True)
- ☐ Radio = NEW radio1("Capital FM")
- ☐ radio1 = NEW Radio("Capital FM")
- ☐ radio1 = NEW Radio("Capital FM", 3, True)

34.2 Radio two

EDUQAS WJEC

isaacacs.org/bq/prog_oop_g02

The following class has been defined using pseudocode.

```

1  CLASS Radio
2      PRIVATE volume: integer
3      PRIVATE station: string
4      PRIVATE on: Boolean
5
6      PUBLIC PROCEDURE Radio(given_station)
7          station = given_station
8          volume = 3
9          on = False
10     ENDPROCEDURE
11
12     PUBLIC FUNCTION get_volume()
13         RETURN volume
14     ENDFUNCTION

```

```
15
16     PUBLIC FUNCTION get_station()
17         RETURN station
18     ENDFUNCTION
19
20     PUBLIC FUNCTION is_on()
21         RETURN on
22     ENDFUNCTION
23
24     PUBLIC PROCEDURE set_volume(new_volume)
25         volume = new_volume
26     ENDPROCEDURE
27
28     PUBLIC PROCEDURE set_station(new_station)
29         station = new_station
30     ENDPROCEDURE
31
32     PUBLIC PROCEDURE switch()
33         IF on == True THEN
34             on = False
35         ELSE
36             on = True
37         ENDIF
38     ENDPROCEDURE
39
40 ENDCLASS
```

► An object has been created based on the class definition shown above and assigned to a reference variable called `my_radio`. Which of the following options (expressed in pseudocode) will set the volume of this radio object to 6?

- ☐ `my_radio.volume = 6`
- ☐ `my_radio.set_volume()= 6`
- ☐ `my_radio.set_volume(6)`

34.3 Radio three

EDUQAS WJEC

isaacss.org/bq/prog_oop_g03

The following class has been defined using pseudocode.

```
1  CLASS Radio
2      PRIVATE volume: integer
3      PRIVATE station: string
4      PRIVATE on: Boolean
5
6      PUBLIC PROCEDURE Radio(given_station)
7          station = given_station
8          volume = 3
9          on = False
10     ENDPROCEDURE
11
12     PUBLIC FUNCTION get_volume()
13         RETURN volume
14     ENDFUNCTION
15
16     PUBLIC FUNCTION get_station()
17         RETURN station
18     ENDFUNCTION
19
20     PUBLIC FUNCTION is_on()
21         RETURN on
22     ENDFUNCTION
23
24     PUBLIC PROCEDURE set_volume(new_volume)
25         volume = new_volume
26     ENDPROCEDURE
27
28     PUBLIC PROCEDURE set_station(new_station)
29         station = new_station
30     ENDPROCEDURE
31
32     PUBLIC PROCEDURE switch()
33         IF on == True THEN
34             on = False
35         ELSE
36             on = True
37         ENDIF
38     ENDPROCEDURE
39
40  ENDClass
```

► In testing, it was found that the volume of the radio could be set to an unsafe level. The `set_volume` method must be updated so that it does not allow the volume to exceed a setting of 30. Order the given statements to create an updated version of the method. You must use all of the statements with correct indentation in your solution.

- ☐ `ENDPROCEDURE`
- ☐ `IF new_volume > 30 THEN`
- ☐ `ELSE`
- ☐ `volume = 30`
- ☐ `PUBLIC PROCEDURE set_volume(new_volume)`
- ☐ `ENDIF`
- ☐ `volume = new_volume`

35 Design

Design is an essential part of application systems development.

Developers need to carefully plan and consider many different solutions before they even start typing code. Every part of a solution needs to be considered, including the **interface** and even the **data** that a program will be working with. Large applications need to be broken down into **modules**, and specific algorithms need to be designed for each important process.

Algorithms can be expressed in a variety of ways. A **flowchart** is a useful visual technique to show the flow of logic through a program. **Pseudocode** is useful when the design will be passed to a programmer who can convert the pseudocode into real program code.

35.1 Match the statement

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/soft_design_g06



► Which of the following statements would you put into the flowchart symbol above?

- ☐ PRINT("Hello")
- ☐ START
- ☐ cost > 10?
- ☐ cost = cost + 25

35.2 Signs of efficiency

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/soft_design_g02

There are often many different algorithms that can accomplish the same task.

► Which of the following characteristics should you consider when comparing the efficiency of two algorithms?

- ☐ The number of comparisons
- ☐ The number of subroutines
- ☐ The number of constants

35.3 What's the output?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/soft_design_g07

Read through the following pseudocode program:

```
1 vowels = ["a", "e", "i", "o", "u"]
2 word = "isaac"
3 count = 0
4
5 FOR EACH letter IN word
6     IF letter IN vowels THEN
7         count = count + 1
8     ENDIF
9 NEXT letter
10
11 PRINT(count)
```

► What will be the output of the program?

35.4 Break it down

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/soft_design_g01

You have been asked to create a program to solve the following problem:

Rik owns a sweet factory, and wants a program to control the temperature of his sweet preparation containers.

Inside each container is a heater. To create his sweets, the mixture of ingredients needs to be heated to a specific ideal temperature.

Rik has containers of different volumes, but it always takes one minute for the heater to raise the temperature of a cubic metre of mixture by one degree.

Rik would like the program to calculate how long the heater needs to be on for to bring the container up to the right temperature.

The different mixtures have different optimum temperatures, so you need to take both this fact and the size of the container into consideration.

► The first step to solving this problem is to break it down into the inputs, processes, and outputs. Use the words below to complete the following decomposition of Rik's problem.

The inputs for this system are the container's _____ temperature and _____, as well as the _____ temperature for the mixture inside. The output is the amount of _____ the heater should be on for. To work out the output, you will need to find the _____ between the two temperatures, then _____ the result by the volume of the container.

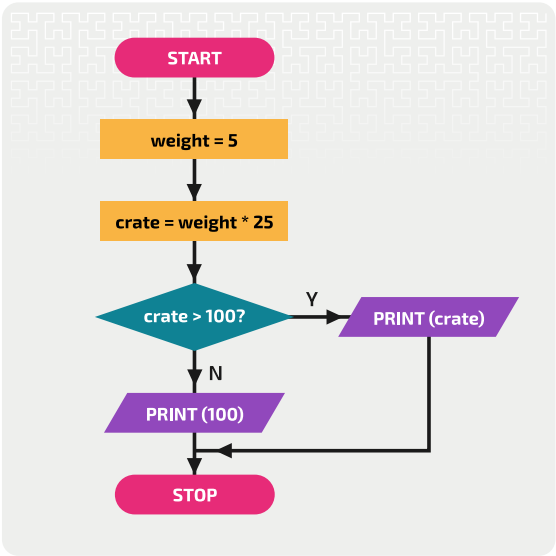
☐ current☐ difference☐ ideal☐ multiply☐ time☐ volume

35.5 Tracing a flowchart

AQA EDEXCEL EDUQAS OCR WJEC

isaacss.org/bq/soft_design_g04

Study the flowchart below that represents an algorithm.



► Trace the algorithm and determine the final output value.

35.6 The final output

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/soft_design_g03

Consider the following pseudocode program:

```
1 number = 4
2
3 FOR count = 1 TO 4
4     PRINT(number * count)
5 NEXT count
```

► What will be the final value output by this program?

Use a trace table like this to help you:

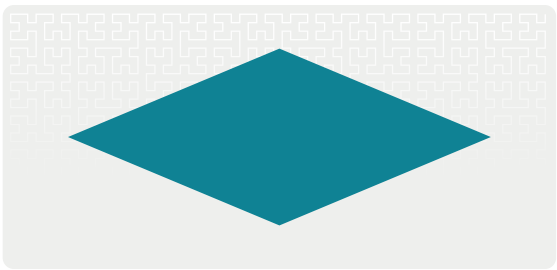
number	count	OUTPUT

Answer:

35.7 Match the statement

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/soft_design_g05



► Which of the following statements could you put into the flowchart symbol shown above?

- ☐ PRINT("Hello")
- ☐ cost = cost + 25
- ☐ cost > 10?
- ☐ START

35.8 Tracing with a table

AQA EDEXCEL EDUQAS OCR WJEC

isaaccs.org/bq/soft_design_g08

Read through the following pseudocode program:

```
1 word = "computer"
2
3 FOR index = 0 TO LEN(word)-1 STEP 2
4     PRINT(word[index])
5 NEXT index
```

► Fill in the trace table by putting the correct values from the options given.

index	OUTPUT
0	
	c

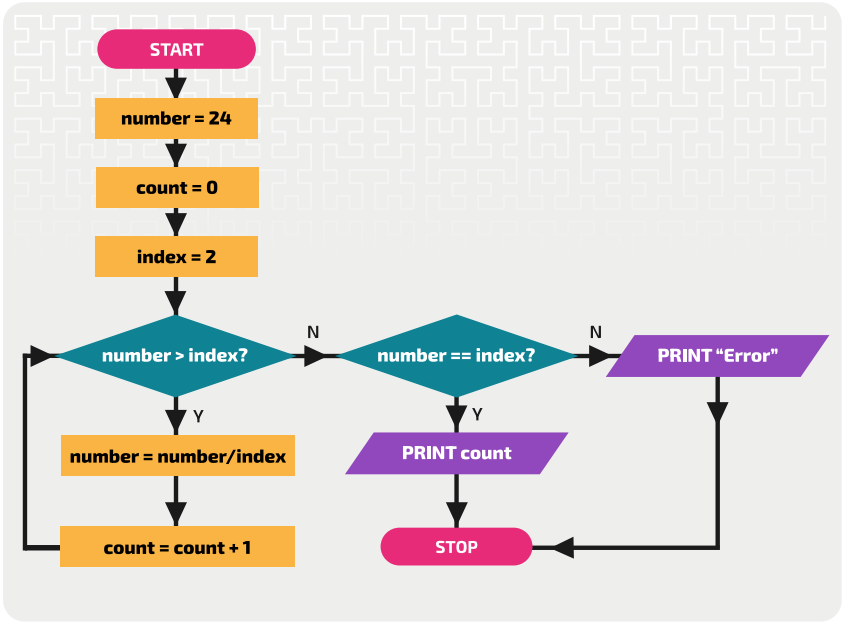
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ o
- ☐ m
- ☐ p
- ☐ u
- ☐ t
- ☐ e
- ☐ r

35.9 Table tracing

AQA EDEXCEL EDUQAS OCR WJEC

isaacscs.org/bq/soft_design_g09

Study the flowchart below that represents an algorithm:



► Trace the algorithm and fill in the trace table by using the correct values from the options given.

number	count	OUTPUT

	0	
12		

	3	
1.5		
	4	

- ☐ 3
- ☐ 4
- ☐ Error
- ☐ 2
- ☐ 6
- ☐ 1
- ☐ 24

36 Testing

Before a piece of software can be used, it needs to be **tested**. Some testing is done by the programmer as the code is written. Other testing is more structured and involves a **test plan** being written and the software being tested with a variety of types of data to check that the outputs are as expected. When a problem with the software is identified, the cause will need to be identified. This is called **debugging**.

36.1 What's the error?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/soft_testing_g02

► The table below contains descriptions of the types of errors that can occur during programming. Add the types of errors into the correct positions in the table.

Description	Type of error
When code runs correctly but the program does not work as expected or produce the expected output.	
When an error occurs during the use of the program. Common errors of this type include trying to divide by 0 or access an item in an array that does not exist.	
When the translator does not recognise the keywords or commands entered. Common errors of this type include mistyping a keyword or missing out brackets.	

- ☐ Syntax
- ☐ Logic
- ☐ Runtime

36.2 On the boundary?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/soft_testing_g01

The program below asks a user to enter a week number and validates their input by checking that the week number entered is between 1 and 52.

```
1 user_input = INPUT("Enter a week of the year: ")
2 week = INT(user_input)
3
4 IF week > 0 AND week <= 52 THEN
5     PRINT("You have chosen week " + STR(week))
6 ELSE
7     PRINT("The week must be between 1 and 52")
8 ENDIF
```

► Which of the options below would be **boundary** test data for this program?

- ☐ -10, -5, 60, and 99
- ☐ 0, 1, 52, and 53
- ☐ 0, 1, 25, and 52

36.3 Week checker

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacacs.org/bq/soft_testing_g04

The program below asks a user to enter a week number and validates their input by checking that the week number entered is between 1 and 52.

```
1 user_input = INPUT("Enter a week of the year: ")
2 week = INT(user_input)
3
4 IF week > 0 AND week <= 52 THEN
5     PRINT("You have chosen week " + STR(week))
6 ELSE
7     PRINT("The week must be between 1 and 52")
8 ENDIF
```

► Write a week number that could be used as a piece of **normal** or **typical** test data for this program.

36.4 Fail to plan, plan to fail

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacss.org/bq/soft_testing_g03

► Below is a testing table that has been developed to test a program that takes in students' test scores and stores them in a database. The system should allow the user to enter the student's name and scores from three separate tests ranging from 0 to 30 marks. Select the correct types of test data from **Boundary, Normal or Erroneous** and place them in the correct positions in the table. Note that the types of test data can be used more than once.

Test ID	Description	Test data	Expected results	Type of test data
01	Check that a student's scores can be added	First name: Billie Surname: Pierce Score 1: 25 Score 2: 20 Score 3: 10	1. A message appears confirming the scores have been added 2. The student's scores are added to the database	
02	Check that only scores between 0 and 30 can be added	First name: Jamie Surname: Morton Score 1: 56 Score 2: 20 Score 3: 10	1. An error message appears asking the user to enter a valid test score 2. No scores are added to the database	
03	Check that only scores between 0 and 30 can be added	First name: Praise Surname: Atwater Score 1: 0 Score 2: 29 Score 3: 30	1. A message appears confirming the scores have been added 2. The student's scores are added to the database	
04	Check that all fields are required fields	First name: Chikere Surname: [leave blank] Score 1: 28 Score 2: 29 Score 3: 27	1. An error message appears asking the user to enter a surname 2. No scores are added to the database	

37 Computational thinking

Computational thinking is the term used to describe some of the skills required in computing, programming, and digital making. Before you can use a computer to solve problems, you need to pick those problems apart and decide how to approach them.

Computational thinking techniques and methods are essential when formulating a problem and expressing a solution so that a computer can effectively it carry out.

Computational thinking covers a broad range of knowledge and skills, including but not limited to:

- Algorithmic thinking – developing an algorithm to solve a problem
- Abstraction – hiding unnecessary detail to reduce complexity
- Automation – taking a model and implementing a solution
- Decomposition – breaking a problem down into smaller parts
- Debugging – detecting and resolving problems in a program
- Logical reasoning – applying rules to solve a problem
- Pattern recognition – identifying the problem as an example of a class of problems that have already been solved
- Problem solving – the way in which solutions are developed

At a more advanced level, you will learn about a wide range of abstraction techniques and computational methods that can be used to help solve complex problems.

37.1 Website woes

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaacscs.org/bq/dsa_ctm_g01

Lottie is developing a website as part of a school project. The website is a large project with multiple pages and she is finding it hard to know where to start!

► Fill in the gaps in the sentence below to explain how Lottie could use decomposition to tackle the problem.

The website project could be _____ into _____, more manageable tasks that Lottie could approach one at a time. These tasks are known as _____.

☐ combined☐ smaller☐ broken down☐ sub-elements☐ larger☐ subtasks

37.2 Which principle?

AQA

EDEXCEL

EDUQAS

OCR

WJEC

isaaccs.org/bq/dsa_ctm_g02

Sami is creating a game. The game will allow the player to move a small alien through a maze to return it safely to its spaceship.

► Sami has started to tackle the problem by hiding unnecessary details; what computational thinking principle has Sami applied?

- ☐ Pattern recognition
- ☐ Algorithmic thinking
- ☐ Abstraction
- ☐ Decomposition

► What **three** pieces of information would be most useful to Sami when developing the initial stages of the game?

At this stage, Sami is focusing on the main features of game play, such as character movement and scoring.

- ☐ What the background of the game will look like
- ☐ What happens if the alien hits the sides of the maze
- ☐ How the score would change if the alien reaches the end of the maze
- ☐ What the next level will look like
- ☐ How the alien will move in the game
- ☐ What music will play during the game

37.3 It is the principle!

AQA EDEXCEL EDUQAS OCR WJEC

isaacs.org/bq/dsa_ctm_g03

► Using the table below, choose the computational thinking principle so that it matches the correct definition.

Definition	Keyword
The process of breaking down large problems into smaller, simpler, more manageable parts	
The process of spotting similarities in smaller problems to develop solutions to more complex problems	
The process of removing unnecessary detail from a problem to aid development of the solution	
The process of identifying the logical steps required to solve a problem	

- ☐ Decomposition
- ☐ Abstraction
- ☐ Pattern recognition
- ☐ Algorithmic thinking